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Master's Thesis

The Spring Sting:

Activity Patterns and Thermal Performance of Halictid Bees on Lesbos Island, Greece

submitted by:

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Acknowledgments

I haven't the faintest idea how to begin this section, and perhaps this is why I've dreaded writing it the most. The two blank pages I've allotted felt cold and daunting for the longest time, because how does one even begin to summarise what feels like an entire lifetime lived within two years? Over these semesters, I did not simply study; I absorbed perspectives, learnt about cultures, breathed in landscapes, and, most importantly, met and shared special memories with people who shaped me in ways I am only beginning to understand now. The friendships and experiences this programme gifted me are far greater than anything that could fit within these pages. This programme taught me how to adapt to unfamiliar places, navigate loneliness and uncertainty, and eventually make islands of my own wherever I went. Even during moments when new places overwhelmed me, there was always comfort in knowing there were people around me who genuinely cared about my wellbeing. My closest friend, David, suggested I contact a friend of his regarding opportunities in ecology on Lesbos. That single suggestion entirely swerved the course of the coming months. David's friend turned out to be Dr Thomas Tscheulin. One meeting with Thomas was enough for me to realise what a privilege it would be to work with him. The trust and freedom he gave me throughout the project shaped my understanding of what mentorship should be. Even while constantly travelling and balancing countless responsibilities, he always found time for me. Beyond being a supervisor, he became a genuine friend, and I consider myself extraordinarily fortunate to have worked alongside him. Through Thomas, I also met Dr Victor Gonzalez and Dr Michael Dillon, both of whom became foundational to this project. Victor's curiosity about the spring activity of *Halictus scabiosae*, combined with his past experience in the region, ultimately gave birth to this study, and he generously shared nesting site information from previous years without which I would not even have known where to begin searching. I am equally grateful to Victor's family — Amy, Oliver, and Rosella — who welcomed me with such warmth that I often felt adopted into their lives. I will always remember the evening walks to Skala Kallonis for ice cream, accompanied by cacophonous frog calls and the occasional surreptitious fox wandering through the dark. Dr Michael Dillon became another invaluable mentor. Beyond lending me equipment that strengthened my fieldwork enormously, he challenged me intellectually and transformed the physiological side of this thesis from something intimidating into something exciting and fun. Many of the best ideas in this project emerged from spontaneous conversations with him. I am grateful to Jelle Devalez for confirming the bee species at my sites and for the many stimulating conversations that followed. I was repeatedly astonished by the depth of his taxonomic knowledge. I also owe sincere thanks to Isabella, whose encouragement and experiences reassured me during moments of uncertainty about my own future. After facing several PhD rejections, the support and belief of the people around me meant more than I can properly express. More broadly, many of the people I met through this project carried perspectives shaped by living in entirely different pockets of the world, and being surrounded by such people constantly expanded the way I viewed both scientific research and life. Finally, I would like to thank Dr Thanasis Kizos for his kindness and generosity throughout my time on Lesbos. The natural history books he lent me greatly helped during the literature review phase and even inspired me during those first difficult stretches of writer's block while beginning the introduction, and his willingness to lend me his action camera quite literally saved a substantial portion of

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The Spring Sting: Thermal Performance and Activity Patterns of Halictid Bees on Lesbos Islands, Greece

Amey Danole^{1,2} and Thomas Tscheulin²

ABSTRACT

Approximately three-quarters of bee species are ground-nesting and spend much of their life cycle underground, where soil microclimate is more important than other macro-environmental variables in shaping their ecology, behaviour, and life history. Despite this, the role of fine-scale thermal heterogeneity during early nest-founding stages remains poorly understood, particularly in primitively eusocial bees. Here, we combine physiological assays and behavioural observations to investigate how microclimatic variation influences thermal performance and activity patterns in foundresses of two ground-nesting halictid bees, *Halictus scabiosae* and *Lasioglossum malachurum*, during spring. First, we measured thermal performance traits (righting and take-off durations and temperatures) to assess species-specific differences in thermal thresholds and recovery capacity. Second, we quantified nest-associated behaviours and general activity at two nesting sites, while recording light, ambient temperature, and soil temperature across a depth gradient. Thermal assays revealed species-specific differences in performance, with *L. malachurum* exhibiting lower righting temperatures than *H. scabiosae*, although flight initiation traits were less differentiated. Behavioural activity was strongly associated with microclimatic conditions, particularly soil temperatures at shallow and intermediate depths, and showed non-linear responses across thermal gradients. Nest guarding behaviour in *H. scabiosae* was influenced by subsurface temperature, highlighting fine-scale environmental sensitivity in behavioural responses. Overall, our findings demonstrate that behavioural activity in ground-nesting halictid bees is shaped by fine-scale microclimatic heterogeneity and reflects species-specific differences in thermal performance. A more complete understanding of these dynamics under environmental change will require integrating stratified soil microclimate measurements, biotic pressures, and individual-level behavioural tracking across sites, seasons, and colony phases.

KEYWORDS: Halictidae, early-season activity, behavioural thermoregulation, chill coma recovery, Aegean islands

INTRODUCTION

Population ecology demonstrates that organisms adapt not only through changes in morphology at the individual level, but also through adjustments in behaviour and energy allocation at the population level (Gould, 1978). Environmental conditions can be both predictable and stochastic, and organisms must cope with this variability by modifying activity patterns and energetic investment in

key fitness-related processes such as growth and development, foraging, and reproduction (Eliopoulos and Stathas, 2003; Whitfield and Richards, 1992), or by changing their distribution (Lomolino et al., 2017; Huang et al., 2022; Davis and Shaw, 2001). Consequently, life-history traits shift across temporal and environmental gradients (Stearns, 1976). Rather than being finely tuned to a single set of conditions, organisms often exhibit phenotypic plasticity that allows them to persist under fluctuating environments (Burton et al., 2022). Such variability generates a broad range of life-history strategies, which are shaped by natural selection and reflect both current ecological conditions and the evolutionary history of the species. One such environmental factor that strongly influences organismal niches and life-history strategies is temperature (Truebano et al., 2018; Gilbert and Raworth, 1996; Howe, 1967; Kamm, 1974; Melville and Schulte, 2001; Porter, 1988). Thermal heterogeneity generates microclimatic variation within habitats, producing fine-scale temperature gradients that can constrain dispersal, influence gene flow, and potentially contribute to population divergence (Johansson et al., 2016). It also influences physiological rates, thereby affecting performance and energy allocation, as well as the nature of biotic interactions, and drives morphological, physiological, and behavioural adaptations that enable organisms to cope with this variability (Tattersall et al., 2012; Kingsolver and Huey, 1998; Abram et al., 2017).

Organisms have evolved ingenious strategies to cope with thermal variation so that key life stages occur when predictable thermal conditions are most favourable (Forrest and Miller-Rushing, 2010). Behavioural adaptations include hibernation in small- and medium-sized mammals, which often follows fat accumulation, as well as in reptiles, which enter prolonged inactivity in response to seasonal conditions. In ectotherms, behavioural adaptations include basking in the sun, shuttling between sun and shade, or adjusting body posture to regulate temperature (Abram et al., 2017). These thermoregulatory behaviours can increase energetic costs, and interactions with conspecifics and predators may further elevate these costs, potentially diverting time and energy from essential activities such as foraging or reproduction. Morphological adaptations may involve reduced insulation in specific body regions to facilitate rapid heat exchange during basking. Physiological adaptations include a low metabolic rate during hibernation and, in species such as ground squirrels, a progressive drop in body temperature to torpor levels and its subsequent recovery during the preparatory phase (Strumwasser, 1960). Additional adaptations include changes in blood pressure and heart rate (Lyman, 1948), convective heat flow associated with high metabolic activity, heat generated during respiration or specialised muscle activity, and complex vascular structures such as counter-current heat exchangers in fish (Cossins, 2012). Some animals, such as moths, alternate between ectothermy and endothermy, being ectothermic while resting and endothermic during flight.

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Ectotherms, characterised by low resting metabolic rates and high thermal conductance, are particularly vulnerable to environmental temperature fluctuations because they rely heavily on external conditions to regulate physiological processes and behaviours (Cossins, 2012). Consequently, a myriad of adaptations—genotypic and phenotypic, including morphological, physiological, and behavioural traits (Bale et al., 2002; Dunham, 1993; Helmuth et al., 2005)—have evolved to cope with temperature variability, whether predictable (e.g., diurnal or seasonal cycles) or stochastic (e.g., extreme weather events). Ground-nesting ectotherms, such as ants, wasps, and bees, occupy the interface between heliothermy (body temperature derived from basking in sunlight) and thigmothermy (body temperature influenced by the thermal properties of their nest environment). This makes it particularly interesting to examine which thermoregulatory strategies they adopt under different conditions to cope with thermal fluctuations. In social insects such as honey bees, thermoregulatory strategies can operate at the colony level, as individuals collectively maintain hive temperatures around 35–36°C across a broad range of external conditions (Cossins, 2012).

The relationship between thermoregulation and social organisation can be further complicated by multiple levels of sociality. Social strategies may be obligate or facultative depending on environmental conditions (Soucy, 2002; Richards et al., 2003). Additionally, multiple broods within a season may exhibit differing strategies depending on their emergence time, highlighting the complexity of behavioural adaptations. Although adult bees from some species are partially endothermic, as they can raise their body temperature using thoracic flight muscles (Willmer and Stone, 2004), ground-nesting bees have limited capacity to regulate nest temperatures directly. Thus, behavioural thermoregulation—through adjustments in activity patterns and the frequency, duration, and timing of behaviours such as nest closing and nest guarding in response to variation in temperature and solar insolation across microclimatic mosaics—emerges as a necessary buffering strategy (Kearney et al., 2009; Bogert, 1949; Bartholomew, 1964), especially in ectotherms that operate close to their thermal maxima (Huey and Stevenson, 1979; Kearney et al., 2009). However, although behavioural thermoregulation via reduced or temporally constrained activity during warm periods may help offset the costs of being active, the resting body temperature and basal metabolic rate of ectotherms remain relatively high (Kearney et al., 2009), meaning that behavioural thermoregulation alone may not be sufficient in the long term.

The sweat bee family Halictidae (Hymenoptera) is a large and taxonomically diverse group of ground-nesting bees (encompassing three quarters of wild bee species (Antoine and Forrest, 2021)) (Knerer, 1969; Fawcett et al., 2019) that exhibit remarkable social polymorphism, spanning a continuum from solitary and communal to semi-social, primitively eusocial, and advanced eusocial forms (Eickworth et al., 1996; Richards, 2000; Dunn and Richards, 2003), with the limits of each category remaining unclearly defined (Antoine and Forrest, 2021). They are present across most faunal zones and consist of closely related species, suggesting ongoing evolutionary diversification within the group (Richards, 1953). However, even closely related species may differ substantially in nest architecture (Knerer, 1969) and, subsequently, in social organisation (Packer and Knerer, 1985; Schwarz et al., 2007), and these species can often be found nesting side by side in the same location (Knerer, 1969; Batra, 1966). Colony sizes range from small (2–4 individuals) to large (over 200), sometimes

with strong division of labour and caste segregation (Lienhard et al., 2010). In some species, multiple foundresses may cooperatively establish a single nest (pleometrosis), after which one female becomes the dominant reproductive individual (Packer, 1986; 1983). This strategy can confer several advantages, including reduced social parasitism, brood parasitism, kleptoparasitism, and nest usurpation (Hogendoorn and Zammit, 2001; Rehan et al., 2011; Dunn and Richards, 2003; Yagi and Hasegawa, 2011; Sick et al., 1994), as well as increased survival, lower nest failure rates, and higher productivity compared to solitary nests (Lin, 1964; Richards and Packer, 1998; Smith et al., 2007). Subordinate females may remain as workers or disperse to establish new nests as colony size increases. Nest reuse across seasons is common, with females often overwintering near their natal nests (Kukuk, 2002). Brood provisioning can influence caste differentiation, determining whether individuals develop into workers or reproductive females that overwinter and initiate colonies in the following season (Kapheim et al., 2011). This diversity in social organisation makes halictid bees a valuable system for examining how sociality interacts with thermoregulatory behaviour. In particular, variation across generations and nesting environments—such as ground-nesting conditions influenced by soil temperature and structure—provides an opportunity to investigate how behavioural and physiological responses to temperature are shaped by both social and environmental factors.

Early-season pollinator activity remains understudied, particularly in ground-nesting bees and in Mediterranean settings (Harmon-Threatt, 2020). Bees are among the most important pollinators globally, and polylectic halictid bees—i.e. generalist foragers that do not specialise on specific host plants but instead utilise a wide range of floral resources—visit diverse wild and cultivated plants and play a crucial role in maintaining ecosystem functioning (Corbet et al., 1993). This study focuses on two primitively eusocial ground-nesting halictid bee species, *Halictus scabiosae* and *Lasioglossum malachurum*, which co-occur on the Greek island of Lesbos (Plateaux-Quénu, 2008; Michener, 2007). These are multivoltine species, producing several generations per year (Michener, 2007; Fawcett et al., 2019), and are polylectic (Shebl, 2012; Bogusch et al., 2020). Bees have a limited capacity to increase heat tolerance through acclimation or prior heat exposure (Gonzalez et al., 2024), making spring-active foundress females particularly vulnerable to rapid temperature increases associated with the ongoing climate crisis. If these reproductive females are adversely affected, subsequent generations may fail to develop, potentially reducing population size and compromising long-term reproductive success and viability. Behavioural thermoregulation may therefore represent a key mechanism by which bees buffer rising temperatures, making it essential to quantify how individuals allocate time to different activities under thermal stress in natural settings.

This study addresses gaps in early-season data by quantifying thermal performance and activity patterns in spring-active foundresses, which may differ from later broods that have been the primary focus of previous research (Gonzalez et al., 2024). By combining behavioural observations with experimentally derived thermal performance metrics, this study links theoretical performance breadth with realised behavioural responses in natural environments. Field observations (at a broad spatial scale) and nesting behaviours (at a focal scale), including nest entry and exit frequencies and nest-guarding duration, are combined with physiological performance measured using a chill-coma recovery assay.

This assay records the time and temperature required for bees to regain righting ability and full flight capacity (take-off), providing two measures of functional recovery from chill-coma impairment. We ask whether variation in recovery performance is reflected in activity and nesting behaviour under variable microclimatic conditions, hypothesising that the less cold-resilient species maintains higher activity levels and greater behavioural persistence during warmer periods of the day and at higher ambient and soil temperatures. Accordingly, we predicted that the less cold-resilient species would recover from chill-coma only at higher temperatures, initiate activity under warmer environmental conditions, and maintain higher abundance and behavioural activity at higher ambient and soil temperatures, while its nesting behaviour would be associated with warmer subsurface soil conditions compared to the other species.

Studying biodiversity patterns across space and time on islands is important (Kier et al., 2009; MacArthur and Wilson, 1967; Schrader et al., 2024; Fritts and Rodda, 1998) because, although islands cover only about 5% of Earth's land area, they harbour approximately 20% of global bee diversity among the roughly 21,000 documented bee species worldwide (Ascher and Pickering, 2020). These bees pollinate the majority of wild flowering plants (Ollerton et al., 2011) and many agricultural crops (Allen-Perkins et al., 2022), while islands also represent important centres of endemism, with 43% of insular bee species restricted to islands (Marshall et al., 2026). Similar to continental patterns, island bee richness peaks at mid-latitudes, and because Lesbos lies only approximately 9 km from the Turkish mainland at its closest point, it supports exceptionally high native bee richness, which generally declines with increasing isolation; furthermore, as a Mediterranean-type ecosystem, it harbours substantially more bee species than native flowering plant species (Marshall et al., 2026). Lesbos represents a particularly suitable system for studying these patterns, as it combines a remarkable diversity of natural ecosystems and agroecosystems within a relatively small area and supports more than 600 recorded bee species (Nielsen et al., 2011). It is considered a global hotspot for bees, with bee abundance ranking second among all Mediterranean islands, surpassed only by Sicily despite the latter being much larger; Lesbos covers only 1,632.8 km² (Dimopoulos et al., 2023; Zevgolis et al., 2022) and is approximately sixteen times smaller than Sicily, yet exhibits exceptional species richness, partly because it draws upon the rich faunal elements of both Europe and Asia (Marshall et al., 2026). This richness has attracted numerous bee studies as well as reflections on the scientific experiences of researchers working on the island (Gonzalez Betancourt, 2023; Roe, 2024). As the third-largest island in the Aegean Sea, Lesbos also possesses an ancient tradition of biological research extending back more than two millennia; Aristotle frequently visited the city of Pyrrha on Lesbos and conducted observations that contributed to his *History of Animals*, regarded by some historians as the first scientific zoological treatise, while he was also among the first to describe the social organisation of honey bees (Tipton, 2006). Likewise, Theophrastus, often regarded as the father of botany, began his studies of plants on Lesbos and subsequently produced two comprehensive botanical treatises, leading the island to be described as the birthplace of biology and, by some accounts, the birthplace of science itself (Gonzalez Betancourt, 2023). Lesbos hosts a unique blend of Anatolian, European, and Mediterranean flora and fauna, lies along a major migratory flyway for birds, and contains a petrified forest dating back approximately 20 million years,

making it a rich natural laboratory for ecological and evolutionary research. Consequently, the island continues to serve as an important study system for diverse topics, including invasive species (Tscheulin et al., 2008), nectar production (Takkis et al., 2015), plant–pollinator interactions (Petanidou and Lamborn, 2005; Potts et al., 2006; Barthell et al., 2009; 2012; Nuttman and Willmer, 2003), beekeeping (Bikos and Rammou, 2002), the effects of landscape parameters on bee body size (Tscheulin et al., 2011) and community assemblages (Neokosmidis et al., 2018), bee foraging patterns (Barthell, 2019), heat tolerance in bees (Gonzalez et al., 2024a; 2020a), bee chronobiology (Cartagena, 2021), and the impacts of climate change on bee communities (Kougioumoutzis et al., 2022), with this work contributing to integrative approaches linking thermal behavioural and physiological responses.

METHODS

Synopsis of bee species

L. malachurum (Kirby, 1802) is a primitively eusocial halictid bee and is widely distributed in the Western Palearctic region across Europe and North Africa and from Iberia to the Caucasus (Polidori et al., 2010; Knerer , 1992; Michener , 1979). *L. malachurum* is approximately 1.7 times smaller than *H. scabiosae* (Rossi, 1790) which is present in the west of the Palearctic and has its primary distribution in western-southern Europe but has recently been recorded in eastern-central Europe (Gil-Tapetado et al. , 2024; Pauly et al. , 2016). Both *L. malachurum* and *H. scabiosae* exhibit flexible, context-dependent social organisation, characteristic of the eusociality gradient in Halictidae, which ranges from solitary to eusocial systems depending on ecological conditions and population structure (Gonzalez et al., 2024a). Both focal species are characterised by long flight seasons from early spring (March) till the start of autumn (September–October) (Gogala , 1991; Knerer , 1980; Weissel et al., 2006; Brand and Chapuisat , 2012). Primitively eusocial species of sweat bees like the two focal species are generally characterised by small colonies, ill-defined castes, frequent queen displacements and a higher proportion of males in the summer brood (Knerer , 1969). *L. malachurum* shows a relatively strict eusocial structure, with a clear reproductive division of labour and temporal separation between brood generations. However, the number of broods produced each season is more flexible, showing clinal variation and tending to increase with higher soil temperatures, although this effect is relatively weak and is also associated with shorter nesting cycles (Weissel et al., 2006). Mated female gynes hibernate during the winter and then found nests solitarily in spring. They give rise to the first brood of about 4–6 daughters, which become workers that assist the long-lived mother with brood cell provisioning and brood rearing, thereby forming a matrilineal society (Knerer , 1969). A second brood consisting of gynes and workers follows, while the third brood is only produced after workers are already present in the colony. However, the number of broods can vary, with populations in warmer climates generally producing more broods per season (Knerer , 1992; Richards , 2000). When the first brood is laid and provisioned for, there is no activity for one to three weeks till their emergence (Weissel et al., 2006). In contrast, *H. scabiosae* exhibits a more flexible form of eusociality with two distinct broods each season. Gynes overwinter in the same nest and, upon emerging from hibernation, reuse it for the following season (Lienhard et al., 2010). At the start of the season, females jointly initiate nests (Gogala , 1991), after which dominance hierarchies form and a single female

typically becomes the primary reproductive individual while the others act as subordinates. The first brood is characterised by a lower proportion of males (Knerer, 1967). The castes can either be well-defined with non-reproductive female workers with undeveloped ovaries (Quénu, 1957) or poorly defined depending on the region (Batra, 1966). The nest-founding female guards the nest while the workers forage (Fabre, 1915) and care for the brood (Knerer, 1980). The semi-social society turns eusocial after the emergence of the first brood of mainly daughters and the other subordinates are chased away by the queen (Gogala, 1991) and may later disperse (drift to nearby areas) to initiate their own nests (Ulrich et al., 2009) or usurp nests of other bees (Westrich, 1990; Knerer and Plateaux-Quénu, 1967). Early-season broods are predominantly female and contribute to colony growth, whereas late-season broods consist of both males and gynes. Males die at the end of the season, while gynes mate in autumn (Gogala, 1991), overwinter and become nest foundresses in the following year.

Study site and design

The study was conducted in an unmanaged pasture near Skala Kallonis (39°12'36"N, 26°12'12"E; 1.0 m a.s.l.) on the Greek island of Lesbos over 10 days (15–24 April 2026), corresponding to spring conditions in the region (Figure 1, Table S8, Table S9, Table S10). Previous work at the same study site has also examined both species in relation to thermal sensitivity (Gonzalez et al., 2020), although during summer conditions. The site consists of bare, sandy alluvial soil with sparse vegetation, providing suitable nesting habitat for ground-nesting bees (Wuellner, 1999; Potts and Willmer, 1997; Wesseling and Tschamtker, 1995; Gregory and Wright, 2005; Severns, 2004). The area also includes patchy anthropogenic disturbance, such as metal fencing, wire, and wooden debris, which is consistent with previous observations that halictine bees may preferentially nest in disturbed habitats (Batra, 1966). Seven managed *Apis mellifera* hives were located approximately 250 m from the study area. Observations were conducted at two sub-sites (hereafter referred to as "sites", approximately 5–6 m²), selected based on high nesting density and clear visibility of nest entrances, which may facilitate nest location via social or environmental cueing (Potts and Willmer, 1997; Michener, 2007). Multi-tunnel nesting aggregations of *H. scabiosae* and *L. malachurum* co-occurred at both sites (Polidori et al., 2010; Antoine and Forrest, 2021; González et al., 2018). The sites were surrounded by an unmanaged wildflower meadow characterised by *Ranunculus sardous*, *Raphanus raphanistrum*, *Trifolium resupinatum*, *Lysimachia arvensis*, and *Matricaria* spp., with additional taxa including *Crepis* spp., *Hordeum* spp., *Rumex* spp., and *Taraxacum* spp.. Nesting aggregations were most frequently associated with bare, sandy patches dominated by sparse *Matricaria* spp., *Trifolium resupinatum*, and *Lysimachia arvensis* vegetation. Six cameras (a heterogeneous system comprising two action cameras, a mobile phone, an iPad, a camcorder, and a DSLR camera) were deployed to record nest-level activity and surrounding arthropod interactions. Two cameras were positioned at Site 1, approximately 1.5 m apart, to capture different perspectives, one wide-angle and one focal view. Three cameras were positioned at Site 2, approximately 1–1.5 m apart, with two capturing focal views and one providing a wide-angle view. An additional camera was positioned between the two sites. However, one camera from Site 2 and the inter-site camera were excluded from the final analysis. The Site

2 camera was excluded due to redundancy with existing recordings, while the inter-site camera was deployed opportunistically to capture behavioural observations rather than systematic sampling. Recordings were conducted for 10 min at the start of each hour. From days 1–6, sampling took place between 10:00 and 18:00, and from day 7 onwards this was extended to 09:00–18:00 to capture earlier onset of daily activity. Individual bees were not marked due to time constraints and to minimise disturbance during recording. In total, 376 10-minute recordings were analysed from four cameras (one wide-angle and one focal camera at each site) and included in the final dataset, corresponding to 188 recordings per site and 94 recordings per camera.

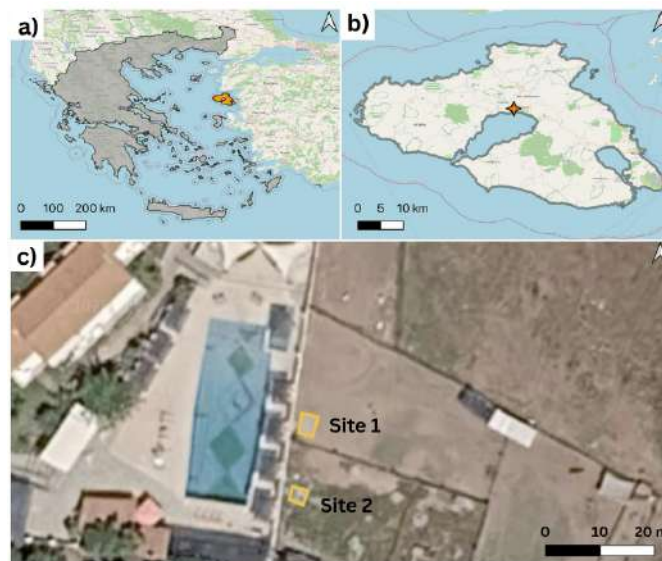


Fig. 1. (a) Geographic location of Lesbos Island highlighted in orange within Greece (grey). (b) Lesbos Island showing the position of Skala Kallonis, indicated by an orange star. Panels (a) and (b) were prepared in QGIS (version 3.40.0 "Bratislava"; QGIS Geographic Information System Development Team). (c) Close-up of the study area in Skala Kallonis, showing the approximate locations of Sites 1 and 2 indicated by yellow borders. Satellite imagery from Google Earth (Google LLC; imagery dated 24 May 2024). The view is shown from an altitude of 107 m above ground level. The midpoint between the two sites is located at 39°12'32.98"N, 26°12'13.74"E, approximately 1 m above sea level.

Chill coma recovery assays

To quantify functional thermal thresholds under field-relevant conditions, we measured the temperature and time required for individuals to recover function following cold immobilisation, defined as (i) righting (regaining upright posture) and (ii) take-off (initiation of flight). Individuals were collected from sites separate from those used for behavioural recordings and placed individually in ventilated plastic vials sealed with cotton (Figure S1). Individuals were not marked for these assays, as sampling across collection sites was sufficient to minimise the likelihood of resampling the same individuals. Chill coma was induced by cooling individuals in either a freezer or a portable cooling box filled with ice and was confirmed by complete immobilisation and a lack of response to gentle mechanical stimulation. Following immobilisation, individuals were transferred to ambient conditions and allowed to recover passively, while ambient temperature was continuously recorded

using a thermocouple thermometer (Omega HH23, type J-K-T), with body temperature monitored simultaneously using a thermal imaging camera (FLIR C5) held above the experimental arena. Body mass was measured prior to trials using a precision balance (Acculab PP2060D; readability 0.001 g), and individuals were handled using fine forceps (Rubis Tech, Switzerland). Intertegular distance (ITD) was measured from dorsal images (see Morphological measurements, Figure S2). During recovery, individuals were positioned ventral-side up to record righting behaviour, defined as the time and thoracic temperature at which a normal upright posture was regained. Take-off temperature was defined as the thoracic temperature at flight initiation. Both righting and take-off temperatures were recorded using the thermal imaging camera. All trials were conducted within a temperature-controlled polystyrene enclosure (26 × 17 cm) to minimise external thermal variation and ensure consistent imaging conditions. The sample sizes collected in this study, based on previous literature (Gonzalez et al., 2024), were considered sufficient to support inferences regarding observed patterns.

Behavioural monitoring and microclimate measurements

Behavioural observations quantified nest-level activity and behavioural repertoires of *H. scabiosae* and *L. malachurum*. Recorded variables included (i) nest entry and exit frequency, and (ii) occurrence and time spent guarding at nest entrances. Nest guarding was quantified only in *H. scabiosae*, as it could not be reliably assessed in *L. malachurum* due to small nesting entrances and restricted camera angles that limited consistent detection of nest-associated behaviour. Consequently, analyses were restricted to *H. scabiosae* (Figure 9, Table S3). Nest aggregation size was estimated visually and assigned to categorical classes (0, <5, 5–10, 10–15, 15–20, 20+ individuals). Wide-angle behavioural recordings were additionally used to estimate relative bee activity at nesting aggregations. Video recordings were sampled at approximately 2, 4, 6, 8, and 10 min following recording onset (± 10 s), and the number of visibly active bees was quantified during each interval. Mean activity was subsequently calculated for each recording session. Abundance was recorded using ordinal categories to generate a relative index of visitation intensity across sites.

Microclimatic conditions were monitored using iButton Thermochron (1-Wire Thermochron; DS1922L; Maxim Integrated, USA) and Hygrochron (1-Wire Hygrochron; DS1923-F5) data loggers deployed across both sites. Temperature loggers recorded air temperature at 2 m height (within a shaded PVC housing) and at multiple soil depths (surface, 2–3 cm, 5 cm, and 10 cm). A hygrometer logger was installed at 10 cm depth near Site 1 to record subsurface humidity and temperature conditions. Nests of *L. malachurum* may reach depths of 10–15 cm during the nest-founding stage (Knerer, 1992; Legewie, 1925; Noll, 1931). Deeper soil measurements (>10 cm) were not conducted in order to minimise repeated disturbance of soil structure, as loggers required periodic retrieval for data verification and maintenance due to battery age and variable reliability. Restricting sampling depth despite nests extending to 20–30 cm depth (Lienhard et al., 2010) reduced disruption to nesting microhabitats and ensured minimal impact on the study system. Light intensity was recorded at each subsite using a smartphone-based lux meter application (Lux Light Meter; Doggo Apps), but these data were excluded from subsequent analyses due to limited explanatory power in preliminary

models. Additional ambient temperature and light intensity data were recorded using HOBO data loggers (Onset Computer Corporation, USA) deployed between Sites 1 and 2 from day 6. Relative humidity was excluded from analyses due to ceiling effects (predominantly 100% saturation, consistent with previous findings (Potts and Willmer, 1997)), which limited variability and interpretability. Surface temperature records from surface-level loggers at Site 1 and Site 2 were missing for 17–18 April 2026 due to a brief technical error.

Morphological measurements

Body size was estimated using ITD as a proxy for body size (Cane, 1987). ITD, defined as the linear distance between the tegulae at the base of the forewings on the dorsal thorax, was measured from dorsal images using ImageJ (Schneider et al., 2012). Each individual was measured twice from independent images, and the mean value was used for analyses. Measurement error was low (mean absolute difference = 0.017 cm; mean coefficient of variation = 1.21%). ITD was analysed using a linear model with species identity as a fixed effect to test for interspecific differences in body size.

Statistical analyses

Linear mixed-effects models (LMMs) were fitted separately for righting duration, righting temperature, take-off duration, and take-off temperature. Fixed effects included ITD, ambient temperature, and species identity in different combinations, with sampling date included as a random intercept. Focal behavioural responses were analysed using generalised linear mixed-effects models (GLMMs), LMMs, and linear models. Nest-guarding occurrence in *Halictus scabiosae* was modelled using a binomial GLMM, and guarding duration using an LMM on log-transformed data. Multi-species abundance was modelled as a function of species identity and scaled environmental variables (ambient temperature, soil temperatures at multiple depths, and light intensity), with site and sampling day included as random effects where appropriate. Wide-angle behavioural responses were analysed using LMMs and generalised additive models (GAMs) to relate overall bee activity to microclimatic conditions. LMMs included temperature across multiple vertical layers and light intensity with site and date as random intercepts, while GAMs were used to capture potential non-linear relationships with environmental predictors. All analyses were performed in R (version 4.4.2; R Core Team, 2024) using RStudio (version 2024.9.1+394; Posit Team, 2024; "Cranberry Hibiscus" release). LMMs were fitted using lme4 (Bates et al., 2015), GLMMs with lmerTest, GAMs with mgcv, and diagnostics were performed using DHARMa. Data visualisation was conducted using ggplot2 (Wickham, 2016).

RESULTS

Interspecific differences in body size

Body size differed between species, with *H. scabiosae* exhibiting a larger ITD than *L. malachurum* (Figure 2, Table S1). Mean ITD was 0.340 ± 0.0329 in *H. scabiosae* and 0.206 ± 0.0220 in *L. malachurum*, corresponding to a 1.65-fold difference in body size between species. A linear model confirmed a significant effect of species identity on ITD, with *L. malachurum* exhibiting a smaller ITD than *H. scabiosae* ($\beta = -0.1340$ relative to *H. scabiosae*). Body mass and ITD were strongly correlated ($r = 0.75$), indicating substantial collinearity between body size metrics. Body

mass was therefore excluded from subsequent analyses, and ITD retained as the primary proxy for body size due to its lower sensitivity to short-term physiological variation.

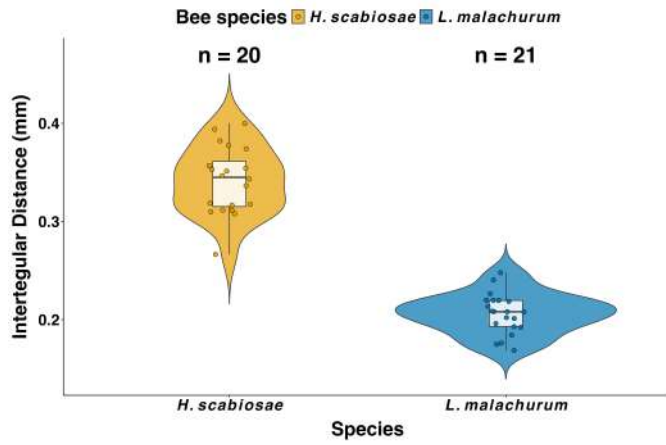


Fig. 2. ITD in *H. scabiosae* and *L. malachurum*. Violin plots illustrate the distribution density of ITD values for each species. Embedded boxplots indicate the median (central line), interquartile range (box), and whiskers extending to $1.5 \times$ the interquartile range; points represent individual observations. Sample sizes (n) are indicated above each group.

Thermal performance

Righting temperature

A linear mixed-effects model indicated a significant positive effect of ambient temperature on righting temperature ($\beta = 0.98$, $SE = 0.15$, $t = 6.554$, $p < 0.001$), indicating that bees righted at higher body temperatures under warmer environmental conditions. Species identity also had a significant effect, with *L. malachurum* exhibiting lower righting temperatures than *H. scabiosae* ($\beta = -2.49$, $SE = 0.88$, $t = -2.841$, $p = 0.0078$), suggesting species-specific differences in acute recovery temperature thresholds (Figure 3). ITD was not included in this final model. Variation among sampling dates was substantial (variance = 1.148), indicating pronounced between-day heterogeneity in righting performance.

Take-off temperature

A linear mixed-effects model indicated a strong positive effect of ambient temperature on take-off temperature ($\beta = 1.41$, $SE = 0.23$, $t = 6.025$, $p < 0.001$), indicating that bees initiated flight at higher body temperatures under warmer environmental conditions (Figure 4). Species identity showed a negative marginal effect, with *L. malachurum* tending to initiate flight at slightly lower temperatures than *H. scabiosae*, although this difference was not statistically supported ($\beta = -2.10$, $SE = 1.12$, $t = -1.884$, $p = 0.069$). ITD had no detectable effect on take-off temperature. Random variation among sampling dates was negligible (variance = 0), indicating minimal temporal heterogeneity in take-off performance. The model showed a boundary (singular) fit, reflecting the absence of meaningful variance at the sampling-date level.

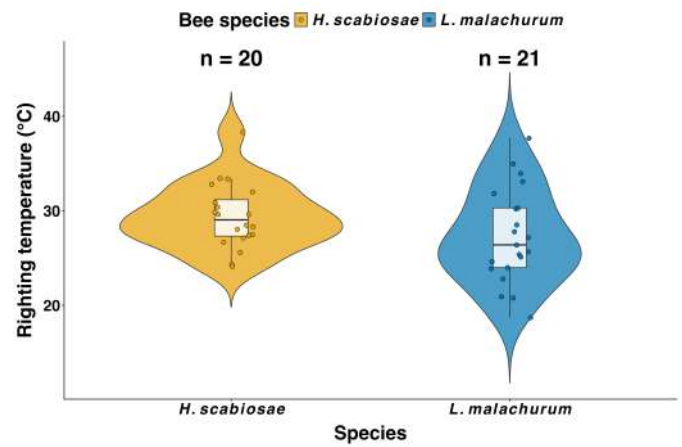


Fig. 3. Righting temperature in *H. scabiosae* and *L. malachurum*. Violin plots illustrate the distribution density of righting temperature values for each species. Embedded boxplots show the median, interquartile range, and whiskers extending to $1.5 \times$ the interquartile range; individual observations are shown as points. Sample sizes (n) are indicated above each group.

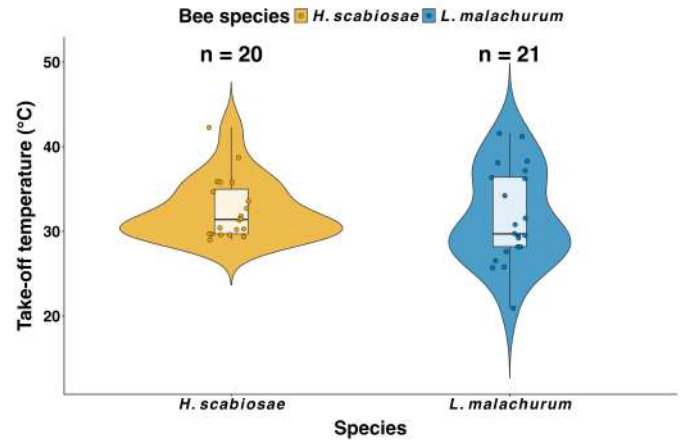


Fig. 4. Take-off temperature in *H. scabiosae* and *L. malachurum*. Violin plots illustrate the distribution density of take-off temperature values for each species. Embedded boxplots show the median, interquartile range, and whiskers extending to $1.5 \times$ the interquartile range; individual observations are shown as points. Sample sizes (n) are indicated above each group.

Behavioural performance

Righting duration

Righting duration (log-transformed) was not significantly affected by ambient temperature. ITD showed a marginal positive effect on righting duration ($\beta = 2.69$, $SE = 1.44$, $t = 1.86$, $p = 0.073$), suggesting a weak tendency for larger individuals to exhibit longer recovery times. Species identity showed a marginal effect, with *L. malachurum* tending to recover faster than *H. scabiosae*, although this difference was not statistically supported ($\beta = -0.42$, $SE = 0.21$, $t = -1.98$, $p = 0.057$; Figure 5). Ambient temperature had no detectable effect on righting duration. Variation among sampling dates was low (variance = 0.032 for the intercept), indicating limited between-day heterogeneity relative to residual variation.

Take-off duration

No significant effects of body size or ambient temperature were detected on take-off duration (log-transformed; Figure 6). ITD had

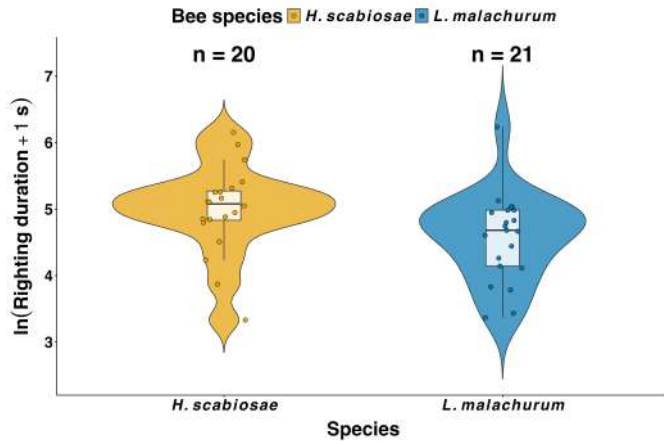


Fig. 5. Righting duration (log-transformed) in *H. scabiosae* and *L. malachurum*. Violin plots illustrate the distribution density of log-transformed righting duration values for each species. Embedded boxplots show the median, interquartile range, and whiskers extending to $1.5 \times$ the interquartile range; individual observations are shown as points. Sample sizes (n) are indicated above each group.

no detectable effect on take-off duration ($\beta = -0.94$, $SE = 1.73$, $t = -0.54$, $p = 0.591$), nor did ambient temperature ($\beta = 0.061$, $SE = 0.053$, $t = 1.15$, $p = 0.259$). Species identity also showed no effect ($\beta = 0.003$, $SE = 0.251$, $t = 0.013$, $p = 0.989$), indicating no evidence for species-specific differences in flight initiation time. Variation among sampling dates was absent, and the model exhibited a singular fit, indicating that between-day heterogeneity did not contribute to variation in take-off duration. Refer to Table S4 for detailed model statistics for physiological and behavioural performance traits in *H. scabiosae* and *L. malachurum*.

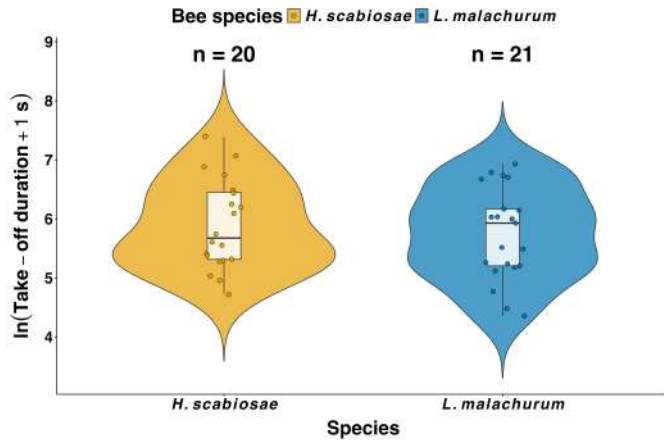


Fig. 6. Take-off duration (log-transformed) in *H. scabiosae* and *L. malachurum*. Violin plots illustrate the distribution density of log-transformed take-off duration values for each species. Embedded boxplots show the median, interquartile range, and whiskers extending to $1.5 \times$ the interquartile range; individual observations are shown as points. Sample sizes (n) are indicated above each group.

Behavioural monitoring and environmental drivers of activity

Bee activity was significantly influenced by microclimatic variation across multiple soil depths and ambient air temperature

(Figure 7, Table S2). Mixed-effects modelling indicated that activity increased with higher ambient air temperatures and warmer shallow-to-intermediate soil temperatures (2–5 cm depth), whereas deeper soil temperatures (10 cm depth) were negatively associated with bee abundance. The model explained a moderate proportion of variation in bee activity (marginal $R^2 = 0.525$; conditional $R^2 = 0.585$), indicating contributions from both fixed effects and temporal variation.

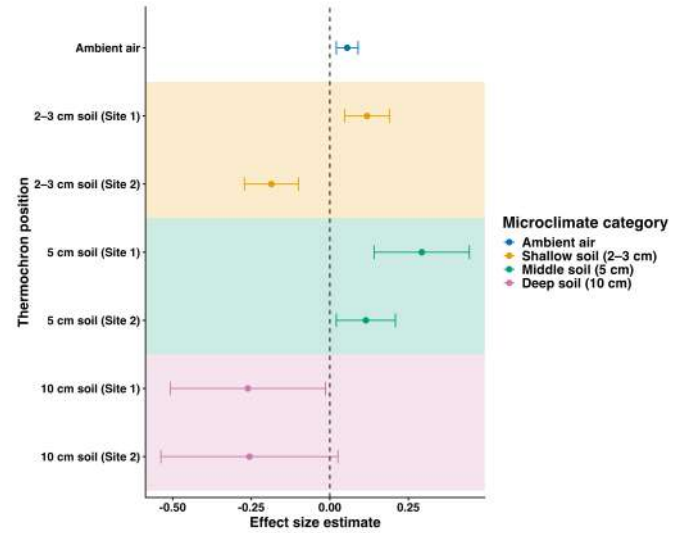


Fig. 7. Linear mixed-effects model estimates of the effects of microclimatic variables at different soil depths and ambient air temperature on mean bee activity (average number of individuals observed). Points represent model estimates and error bars indicate standard errors.

Generalised additive models (Figure 8) further demonstrated that these relationships were non-linear across thermal gradients. Bee activity showed a significant non-linear response to ambient air temperature ($edf = 3.19$, $F = 18.98$, $p < 0.001$), explaining 43.9% of deviance. Intermediate soil temperature (5 cm depth) and deep soil temperature (10 cm depth) also exhibited significant non-linear effects ($p = 0.007$ and $p < 0.001$, respectively), whereas shallow soil temperature (2–3 cm depth) was not significant. However, these results should be interpreted with caution, as the focal observations do not allow individual bees to be reliably distinguished. Consequently, the possibility of pseudoreplication cannot be excluded, as the same individual may have been recorded multiple times across the same observation period.

Nest guarding duration and probability of nest guarding in *H. scabiosae*

Nest guarding duration increased significantly with soil temperature at 5 cm depth ($\beta = 0.356 \pm 0.139$, $t = 2.55$, $p = 0.016$), whereas ambient air temperature, surface temperature, shallow soil temperature, and deep soil temperature were not significant predictors. Random effects indicated moderate variation among days and low variation among sites.

In contrast, the probability of nest guarding decreased significantly with increasing shallow soil temperature (2–3 cm depth; $\beta = -0.94 \pm 0.27$, $z = -3.50$, $p < 0.001$), indicating reduced guarding activity under warmer subsurface conditions. Variation among days and sites was moderate.

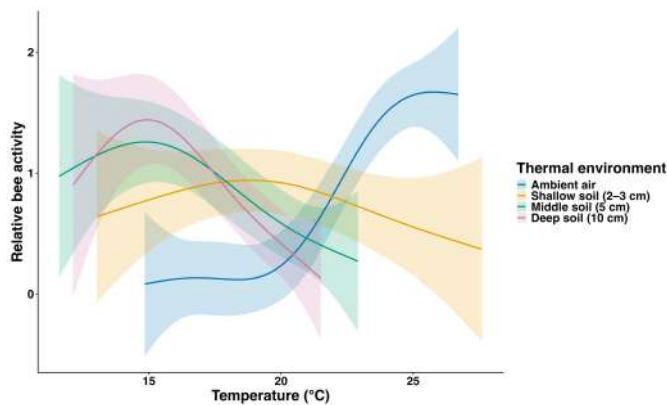


Fig. 8. Generalised additive model (GAM) fits showing non-linear relationships between bee activity and thermal variables across microclimatic gradients. Solid lines represent fitted smooth functions, with shaded areas indicating 95% confidence intervals. Panels show ambient air temperature and soil temperatures at different depths (2–3 cm, 5 cm, and 10 cm).

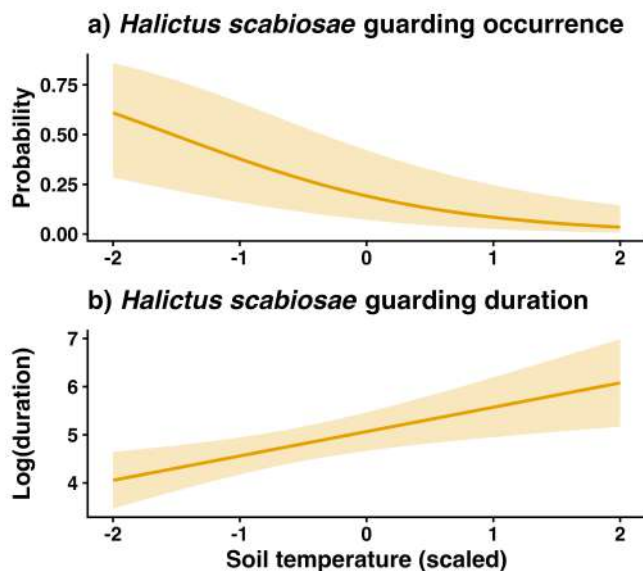


Fig. 9. Effects of subsurface soil temperature on nest guarding behaviour in *H. scabiosae*. (a) Predicted probability of nest guarding occurrence as a function of shallow soil temperature (2–3 cm depth) derived from a binomial generalised linear mixed-effects model. (b) Predicted nest guarding duration (log-transformed) as a function of middle soil temperature (7–8 cm depth) derived from a linear mixed-effects model. In both panels, soil temperature was standardised prior to modelling, such that values of -2 and $+2$ represent approximately ± 2 standard deviations from the mean observed temperature at each respective depth. This corresponds to the following raw temperature ranges: shallow soil (mean $\pm$ SD: 19.65 ± 3.20 °C; range: 13.26 – 26.05 °C) and middle soil (18.51 ± 2.83 °C; range: 12.84 – 24.18 °C). Solid lines represent model predictions and shaded areas indicate 95% confidence intervals.

Species differences in abundance and environmental responses

Bee abundance (log-transformed) was analysed using a linear model including species identity, environmental variables, and species-by-environment interaction terms (Figure 10). The model explained a substantial proportion of variation in abundance

($R^2 = 0.556$; $F_{11,178} = 20.3$, $p < 0.001$). *L. malachurum* exhibited significantly lower overall abundance than *H. scabiosae* ($\beta = -0.368 \pm 0.051$, $t = -7.25$, $p < 0.001$). Abundance increased significantly with shallow soil temperature (2–3 cm depth; $\beta = 0.318 \pm 0.083$, $t = 3.85$, $p < 0.001$) and light intensity ($\beta = 0.242 \pm 0.051$, $t = 4.74$, $p < 0.001$), whereas middle soil temperature (5 cm depth) showed a significant negative association with abundance ($\beta = -0.289 \pm 0.103$, $t = -2.81$, $p = 0.006$). Deep soil temperature did not significantly predict abundance. Significant species-by-environment interactions were detected for ambient air temperature ($\beta = 0.088 \pm 0.079$, $F = 13.82$, $p < 0.001$) and light intensity ($\beta = -0.207 \pm 0.072$, $t = -2.87$, $p = 0.005$), indicating species-specific environmental responses. Specifically, *L. malachurum* exhibited weaker positive responses to increasing light intensity relative to *H. scabiosae*. Interactions with shallow, middle, and deep soil temperatures were not statistically supported.

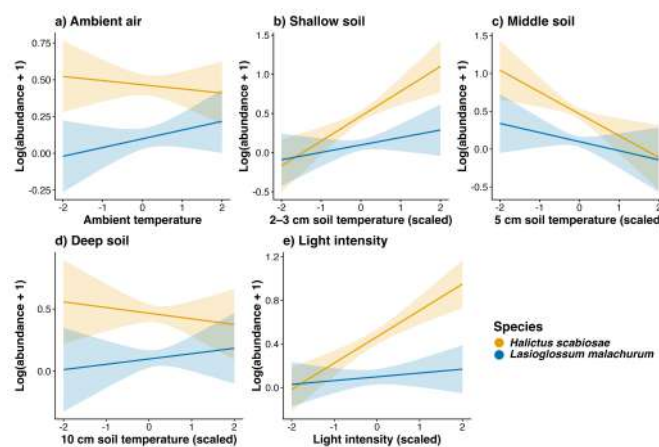


Fig. 10. Relationships between environmental variables and bee abundance in *H. scabiosae* and *L. malachurum*. Panels show predicted abundance responses across (a) ambient air temperature, (b) shallow soil temperature (2–3 cm depth), (c) middle soil temperature (7–8 cm depth), (d) deep soil temperature (10 cm depth), and (e) light intensity. Soil temperature variables were standardised prior to modelling; therefore, values of -2 and $+2$ represent approximately ± 2 standard deviations from the mean observed temperature at each depth. This corresponds to shallow soil temperatures of 19.65 ± 3.20 °C (13.26 – 26.05 °C), middle soil temperatures of 18.51 ± 2.83 °C (12.84 – 24.18 °C), and deep soil temperatures of 17.62 ± 2.17 °C (13.27 – 21.96 °C). Solid lines represent model predictions and shaded ribbons indicate 95% confidence intervals.

Environmental correlates of movement frequency

Generalised additive models revealed species-specific relationships between movement frequency and environmental conditions (Supplementary Figures S3–S9; for model statistics refer to Table S5). Nest entry frequency in both *H. scabiosae* and *L. malachurum* exhibited significant non-linear smooth relationships with shallow soil temperature (*H. scabiosae*: $\text{edf} = 2.14$, $F = 3.54$, $p = 0.024$; *L. malachurum*: $\text{edf} = 2.36$, $F = 6.32$, $p < 0.001$). Nest exit frequency in *H. scabiosae* was significantly associated with shallow soil temperature ($F = 4.46$, $p = 0.038$), whereas outgoing flight frequency in *L. malachurum* was significantly associated with ambient air temperature ($F = 4.61$, $p = 0.035$) and showed a marginal non-linear relationship with middle soil temperature ($\text{edf} = 3.16$, $F = 2.44$, $p = 0.052$). Light intensity and deep soil

temperature were generally not significant predictors across models. Overall, the GAMs explained between 23.0% and 40.5% of deviance across models.

DISCUSSION

In this study, we investigated species-specific patterns of abundance and behavioural responses to environmental variation in temperature and light intensity, alongside thermal performance measured via chill coma recovery, in two sweat bee species, *H. scabiosae* and *L. malachurum*, on Lesbos island, Greece. Body size differed significantly between species, with *H. scabiosae* exhibiting larger ITD than *L. malachurum*. Given established positive relationships between body size and foraging range, this difference may imply interspecific variation in potential foraging distances (Greenleaf et al., 2007; Gathmann et al., 1994), which could in turn influence access to spatially distributed floral resources.

Chill-coma recovery assays

Chill-coma recovery assays revealed a consistent effect of ambient environmental temperature on both righting and take-off temperatures in *H. scabiosae* and *L. malachurum*. In both assays, bees initiated recovery and flight at higher body temperatures under warmer ambient conditions, indicating that behavioural thermal thresholds shift in response to prevailing environmental conditions. Species identity significantly influenced righting temperature, with *L. malachurum* exhibiting lower recovery temperatures than *H. scabiosae*, whereas for take-off temperature this difference was weaker and not statistically supported. ITD did not explain variation in either thermal threshold. In contrast, behavioural recovery durations showed weaker structure overall: righting duration displayed only marginal effects of body size and species identity, whereas take-off duration showed no detectable effects of any predictor. Variation among sampling dates was present for righting temperature but negligible for take-off temperature and both behavioural duration metrics. Overall, acute thermal thresholds showed partial species-specific differentiation, whereas behavioural recovery times were largely insensitive to species identity and body size. Individuals were not marked for these assays, as sufficient sampling across sites allowed individuals to be drawn from a large pool, reducing the likelihood of resampling the same individuals.

Species-specific behavioural responses to microclimatic variation

H. scabiosae was substantially more abundant than *L. malachurum* across focal observations, indicating clear differences in observed activity patterns between species. However, this pattern may partly reflect differences in nest detectability and sampling bias, as *H. scabiosae* nests were larger and more visually conspicuous than those of *L. malachurum*, potentially resulting in greater representation within behavioural recordings. Differences in local nest abundance between species may have also contributed to the observed disparity in activity levels. Behavioural activity in both species was strongly influenced by thermal and light conditions, highlighting the importance of microclimatic variation in structuring above-ground activity. Overall, bee abundance increased with higher ambient air temperatures (Lienhard et al., 2010), warmer shallow soil conditions, and increased light intensity. These environmental

variables are partially correlated and collectively represent conditions that facilitate foraging activity by reducing thermoregulatory constraints and improving flight performance and activity efficiency (Willmer and Stone, 2004). Species-specific responses further revealed distinct environmental coupling. *L. malachurum* showed a stronger association between activity and ambient air temperature, particularly in outgoing flight frequency, suggesting greater sensitivity of foraging-related movement to above-ground thermal conditions. In contrast, *H. scabiosae* exhibited comparatively stronger associations with subsurface thermal conditions, indicating a closer link between nest-adjacent microclimatic variation and behavioural expression. Nest guarding in *H. scabiosae* showed two distinct environmental responses. The probability of nest guarding decreased with increasing shallow soil temperature, indicating reduced likelihood of engaging in guarding behaviour under warmer subsurface conditions. In contrast, guarding duration increased with soil temperature at intermediate depth, suggesting that once guarding was initiated, warmer nest-adjacent conditions were associated with longer behavioural bouts. Nest entry and exit frequencies (used as proxies for nest-associated activity and foraging activity, respectively) exhibited species-specific responses to environmental variation. *L. malachurum* showed a stronger association between ambient air temperature and outgoing nest flights, indicating increased responsiveness of foraging activity (as inferred from nest exit frequency) to above-ground thermal conditions. In contrast, *H. scabiosae* displayed stronger links between nest entry and exit frequencies and subsurface thermal conditions, suggesting that nest traffic is more closely coupled to microclimatic variation within or immediately surrounding the nest environment. Light intensity showed a weaker and less consistent effect on nest entry and exit frequencies in both species. Because soil temperatures across depths are inherently correlated, coefficient estimates should be interpreted as indicators of relative associations across the thermal profile rather than fully independent thermal effects.

Decoupling of behavioural activity and physiological thermal responses

Behavioural activity patterns and physiological thermal responses showed only partial alignment in their sensitivity to ambient temperature, suggesting that thermal effects operate differently across levels of biological organisation. Although warmer conditions were associated with increased behavioural expression and elevated body temperatures during recovery and flight initiation, the inconsistency in response strength indicates that behavioural output is not tightly constrained by physiological thermal limits. Behavioural variation appeared strongly species-dependent and closely linked to microclimatic context, particularly in nest-associated movement and foraging activity, implying that ecological context may mediate behavioural thermoregulation more strongly than intrinsic physiological thresholds. In contrast, physiological thermal responses showed weaker and less consistent species differences, with divergence evident in righting temperature but not in take-off temperature, and no detectable effect of body size, suggesting limited physiological differentiation between species. Behavioural recovery durations were largely insensitive to both species identity and environmental temperature, further supporting a decoupling between acute thermal physiology and behavioural kinetics. Overall, these findings indicate that while behaviour and physiology respond to temperature in

broadly similar directions, behavioural performance is not directly predicted by physiological thermal thresholds, highlighting a functional disconnect between organismal-level activity patterns and underlying thermal physiology.

Thermal ecology and nesting behaviour in halictid bees

Halictidae display substantial variation in social organisation within and among species (Hirata et al., 2005; Packer, 1998; Plateaux-Quénu, 1993; Richards et al., 2003; Yanega, 1988), which is often associated with environmental conditions (Yanega, 1997; Eickwort et al., 1996; Miyanaga et al., 1999) and, in some cases, genetic influences on social traits (Danforth et al., 2003; Plateaux-Quénu et al., 2000; Soucy, 2002). Ground-nesting insects are more strongly influenced by soil temperature than air temperature in relation to nest-site selection, daily activity patterns, and energetic allocation to foraging and life-history processes (Cameron et al., 1996; Crist and Williams, 1999; Potts and Willmer, 1997; Strohm and Linsenmair, 1998; Vogt et al., 2003; Weissel et al., 2006). In *L. malachurum*, increasing soil temperatures reduce brood development time and shorten activity pauses between broods, likely due to physiological effects on larval development (Weissel et al., 2006). The higher activity levels observed at warmer temperatures in previous studies are consistent with patterns observed here (Weissel et al., 2006). For *H. scabiosae*, nesting preferences could be examined in a similar framework to *Halictus rubicundus*, which shows higher nesting densities in areas with warmer soil temperatures at 5 cm depth relative to surrounding soils, consistent with brood cell depth (Potts and Willmer, 1997). Because surface and shallow soil temperatures are correlated (West, 1952; Xu et al., 1997; Rajver et al., 2006), they may provide reliable proxies of subsurface thermal conditions, potentially allowing females to use thermal cues when selecting nest sites. Exposure to sublethal desiccation stress did not alter CT_{max} (maximum critical temperature) or time to heat stupor in *L. malachurum* (Gonzalez et al., 2024a). Similarly, prior starvation followed by sublethal desiccation did not shift mean CT_{max} , but increased variance in CT_{max} , suggesting greater inter-individual variability in thermal tolerance (Gonzalez et al., 2024a). If similar variability is present among foundress females, it may influence which individuals successfully establish subsequent generations under acute thermal stress, with potential consequences for population dynamics.

Future directions

Personal observations during fieldwork and analysis of behavioural videos revealed that multiple individuals from the same species may share a common nest entrance. This warrants further investigation to determine whether multiple foundresses contribute to nest establishment or whether individuals merely share entrance access, as observed in other halictid bee species (Paxton et al., 1999). Marking of individuals would be required to resolve this and to test whether dominant females exhibit aggressive nest-guarding behaviour to the extent that even returning nestmates may be excluded, as suggested by (Lienhard et al., 2010; Knerer, 1980; Knerer and Plateaux-Quénu, 1967). Field observations also indicated the presence of velvet ants (Mutillidae), parasitic wasps (Hymenoptera), and bombyliid flies (Bombyliidae) (Batra, 1966; Lienhard et al., 2010) in the vicinity of nesting aggregations. These individuals were frequently observed inspecting nesting sites, particularly during afternoon periods when nests were often

closed. Characterising brood parasite presence and activity patterns may therefore provide additional insight into nest-associated behavioural dynamics in halictid bees in future studies (Sick et al., 1994; Weislo and Cane, 1996). Nest opening (Figure S10) and closing (Figure S11) durations and timings (Figure S12), and the environmental conditions under which these occur, should also be recorded for these species, as this may provide insight into thermal tolerance and the potential influence of predators or nest-infesting parasites on nest-closure behaviour and activity patterns (Lienhard et al., 2010; Sih and McCarthy, 2002; Houston et al., 1993), as well as changes in energy budget expenditure associated with differential predation and parasitism (Minchella, 1985; Vandame et al., 2002). In particular, determining whether nest closure is primarily driven by temperature or by the presence of parasites or predators could improve our understanding of nest defence strategies in ground-nesting bees (Wuellner and Jang, 1996). For the current study, nest-opening and closing duration could only be observed for *H. scabiosae* nests since they were clearly visible. Nest-opening duration varied from 2–118 s (mean $\pm$ SD: 32.2 ± 42.0 s; $n = 6$), while nest-closing duration varied from 15–456 s (mean $\pm$ SD: 170.0 ± 162.6 s; $n = 5$). Future studies should also quantify burrowing and nest-switching behaviour, which were not captured in the present study. Mark-recapture approaches could be used to determine how frequently individuals change nesting sites within aggregations. Preliminary field observations suggest that both *H. scabiosae* and *L. malachurum* may inspect and enter multiple nests within the same aggregation, particularly early in the nesting cycle, potentially facilitating nest switching or usurpation of unguarded nests, which appears to be more frequent in *H. scabiosae* (Knerer, 1969). However, this behaviour requires systematic quantification, particularly given that nesting architecture differs between species, with *H. scabiosae* favouring short laterals and sessile cells, whereas *L. malachurum* constructs brood cells at the end of more extensive lateral tunnels (Knerer, 1969). Finer-scale life-history strategies are more directly determined by intrinsic colony factors such as genetic composition, queen quality, and local resource distribution, which ultimately govern colony dynamics (Weissel et al., 2006), while broader-scale environmental characteristics provide the ecological context within which these processes operate. Consequently, comparative genetic analyses across colonies within and between regions and environmental conditions are required to assess the extent to which fine-scale population structure underlies variation in these colony-level processes.

Nesting habitat parameters such as soil type, texture, compaction, pH, aeration, bulk density, and moisture, which influence thermal diffusivity (Marshall et al., 1996), as well as microhabitat preferences including nest entrance orientation, vegetation cover, slope, shade, and nearby surface stones (Jones and Oldroyd, 2006; Antoine and Forrest, 2021; Packer et al., 1989; Packer and Knerer, 1986b; Cane, 2015; Potts and Willmer, 1997; Shebl et al., 2016), and exposure to temporally variable insolation (e.g. morning sun rather than total daily insolation; (Sakagami and Hayashida, 1961)), remain poorly understood. How these factors influence bee fitness warrants further investigation (Antoine and Forrest, 2021). Furthermore, because such nesting preferences cannot be generalised across all ground-nesting bees (Harmon-Threatt, 2020), population-level studies are needed to assess inter-population variation in nesting ecology and microhabitat use. The initial days of the nest-founding stage in early spring, when weather conditions are unpredictable and soil temperatures are relatively colder

than in summer, are often poorly characterised (Weissel et al., 2006) and represent a key gap that future studies should address within the nesting cycle. Increasing soil temperature may also accelerate ageing in workers and increase developmental and survival rates of larvae (Cossins, 2012; Jeanne and Morgan, 1992; Whitfield and Richards, 1992), and thus lead to shorter nesting cycles with reduced foraging durations and fewer pauses between broods (Weissel et al., 2006). Previous studies have shown that soil temperature during the nest-founding to first brood stage in *L. malachurum* may be more influential than summer soil temperatures in determining colony development (Weissel et al., 2006). Based on a global review of 449 species of ground-nesting bees, the average minimum nesting depth is approximately 17 cm, with variation linked to brood stage and nesting cycle (Weissel et al., 2006). This suggests that the 10 cm maximum soil depth sampled in the present study may not fully capture deeper nesting microclimatic conditions, and that finer thermal gradients at greater depths could provide additional ecological insight (see Cane and Neff, 2011). Furthermore, in *L. malachurum*, colder soil temperatures at depths of 20–40 cm have been shown to slow nest construction and founding rates (Weissel et al., 2006), indicating that species-specific nesting depths should be considered when assessing thermal effects on colony development. Also, since the field sites in this study were situated only 7–10 meters apart, future studies could examine sites in different locations and compare activity patterns and behaviours of the same species to better resolve site-specific effects (see Stone, 1994; Redding et al., 2003; Davis et al., 2019; Mikola et al., 2018). Personal observations indicated the presence of turret-like soil structures at some nest entrances, while others lacked such structures. Beetles and orthopterans, including grasshoppers and pygmy grasshoppers (Tetrigidae), were frequently observed basking on these turrets and appeared to be tolerated by resident bees. Future studies could compare nests with and without turrets, as elevated structures may heat more rapidly than surrounding soil and influence nest entrance microclimate (see Norden, 1984).

Thermal limits and responses to climate change in bees

Temperature strongly influences the physiology, behaviour, mating (Larsson, 1991), energetic costs (Kearney et al., 2009), and life-history strategies such as timing of adult emergence (Forrest and Chisholm, 2017), number of generations per year (Forrest et al., 2019), and larval maturation and development in insects (Borrell and Medeiros, 2004; Grafton-Cardwell et al., 2005; Woods et al., 2005). Warmer temperatures can lead to shorter nesting cycles and a higher number of generations per year, but may also result in smaller body size and increased mortality in ectotherms; thus, population dynamics and fitness consequences are highly complex and context-dependent (Kingsolver and Huey, 2008; Forrest and Chisholm, 2017). Alongside rising temperatures, a suite of anthropogenic stressors—including habitat loss, pesticide exposure, and environmental disturbances such as wildfires (Cane and Neff, 2011), whose frequency is increasing, shifts in vegetation cover driven by elevated atmospheric CO₂ (Hoffmann et al., 2002), flooding events (Roubik and Michener, 1980; Potts et al., 2016), reduced floral resources due to invasive species, and pathogen and parasite spillover from managed bees—represent major contemporary threats to bees. Maintaining and restoring native vegetation can provide thermal refuges during heat stress or heat waves. Under such anthropogenic pressures, as observed at the study

site, conditions for ground-nesting bees are highly variable, with microhabitat temperatures subject to stochastic fluctuations; consequently, the ability to cope with and adapt life-history strategies to these conditions is essential for survival and reproduction. *H. scabiosae* has recently been reported to expand its range into previously unsuitable habitats in response to environmental change, with urbanisation playing a major role in this process; however, its ecological niche—characterised by high temperature seasonality and low precipitation—has remained relatively stable from 1970 to the present, suggesting limited adaptability to novel conditions (Gil-Tapetado et al., 2024). Consequently, habitat fragmentation and increasing urban density (Lanner et al., 2022; Polidori et al., 2021) may facilitate colonisation of previously unoccupied areas by *H. scabiosae* (Gil-Tapetado et al., 2024), although not all bee species, particularly those with specialised resource use and limited dispersal capacity, can track suitable climates by shifting poleward or to higher latitudes (Kerr et al., 2015; Martínez-López et al., 2021; Casey et al., 2015; Nemésio et al., 2016; Buckner and Danforth, 2022). During the sampling year, extreme rainfall events occurred, with maximum daily precipitation reaching 49.2 mm, within a system characterised by highly variable inter-annual rainfall (21.2–83.9 mm annually across 2003–2019; Agia Paraskevi meteorological station, meteo.aegean.gr), typical of the Mediterranean climate of Lesbos. The progressively earlier onset of spring in recent years highlights how shifting environmental conditions may require bees to adjust the timing of seasonal activity and termination (Kearney et al., 2009) and adopt novel behavioural strategies to cope with thermal and climatic variability. Over evolutionary timescales, species have shifted their geographic distributions, and under ongoing climate change such shifts are expected to become more frequent as previously suitable habitats become inhospitable. However, physiological heat tolerance is often conserved across lineages and may be constrained by relatively fixed upper thermal limits beyond which performance cannot be maintained (Araújo et al., 2013; Hoffmann et al., 2013). In this context, CT_{max} values of *H. scabiosae* (43 °C) and *L. malachurum* (42 °C) (Gonzalez et al., 2024) represent important components of their thermal niches (Araújo et al., 2013), yet remain above the mean maximum summer air temperatures recorded on Lesbos. Specifically, mean summer maximum temperatures at the Agia Paraskevi meteorological station (Meteo Aegean; nearest long-term station to Skala Kallonis) were 28.0 °C over the period 2003–2019, based on July–August maxima (and June where applicable; see Supplementary Table S10).

Across ectotherms, thermal performance typically peaks between 30–35 °C, with heat stress occurring above 40 °C (Deutsch et al., 2008; Addo-Bediako et al., 2000; Frazier et al., 2006). Consequently, although ambient summer temperatures at the study site remain below CT_{max}, they approach ecologically relevant thermal performance thresholds, placing individuals close to their thermal safety margins. Under such conditions, populations may become increasingly vulnerable to extreme heat events and climate warming (see Deutsch et al., 2008; Huey et al., 2009; Clusella-Trullas et al., 2011) and may experience shifts in their distributions (Chen et al., 2011). Behavioural thermoregulation may therefore buffer individuals against exceeding physiological limits and increase short-term persistence under thermal stress. However, despite its plasticity, behavioural buffering is ultimately constrained by physiological limits (Huey et al., 2003), limiting the extent to which realised niches can expand beyond conserved fundamental thermal boundaries (Jackson and Overpeck, 2000). This

operates within a broader thermodynamic niche framework (Kearney et al., 2013), shaped by dispersal limitation (Baselga et al., 2012), historical contingency (Ricklefs, 2006), and biotic interactions (Dobzhansky, 1950), which together determine realised ecological distributions (Araújo et al., 2013). Although these processes are often highly labile, they impose constraints that ultimately set limits to evolutionary responses (Huey et al., 2003). Consequently, the realised niche is constrained by conserved fundamental niche properties across lineages (Jackson and Overpeck, 2000), and the so-called "thermodynamic niche" represents the stage on which ecological and evolutionary interactions unfold (Kearney et al., 2013). However, the extent to which realised niche differences reflect dispersal tendencies (Baselga et al., 2012), historical biogeographical legacies (Ricklefs, 2006), rates of environmental change, resource utilisation, and biotic interactions (Dobzhansky, 1950) remains to be comprehensively explored (Araújo et al., 2013).

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Contribution

Amey Danole designed and conducted the study, performed fieldwork, analysed the data, and wrote the manuscript. Dr Thomas Tscheulin supervised the project, contributed to the conceptualisation of the behavioural study, provided materials for behavioural observations, and assisted with manuscript revision.

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Data availability

The data supporting the findings of this study are openly available in the following GitHub repository: https://github.com/amey-danole/Masters_Thesis_Lesvos.

REFERENCES

Abram, P. K., Boivin, G., Moiroux, J. and Brodeur, J. (2017). Behavioural effects of temperature on ectothermic animals: unifying thermal physiology and behavioural plasticity. *Biol. Rev.* 92, 1859–1876.

Addo-Bediako, A., Chown, S. L. and Gaston, K. J. (2000). Thermal tolerance, climatic variability and latitude. *Proc. Biol. Sci.* 267, 739–745.

Allen-Perkins, A. et al. (2022). CropPol: A dynamic, open and global database on crop pollination. *Ecology* 103, e3614.

Antoine, C. M. and Forrest, J. R. K. (2021). Nesting habitat of ground-nesting bees: a review. *Ecol. Entomol.* 46, 143–159.

Araújo, M. B., Ferri-Yáñez, F., Bozinovic, F., Marquet, P. A., Valladares, F. and Chown, S. L. (2013). Heat freezes niche evolution. *Ecol. Lett.* 16, 1206–1219.

Ascher, J. S. and Pickering, J. (2020). Discover Life bee species guide and world checklist (Hymenoptera: Apoidea: Anthophila). *Discover Life*. Available at: http://www.discoverlife.org/mp/20q?guide=Apoidea_species.

Bale, J. S. et al. (2002). Herbivory in global climate change research: Direct effects of rising temperature on insect herbivores. *Glob. Change Biol.* 8, 1–16.

Barthell, J. F., Clement, M. L., Song, D. S., Savitsky, A. N., Hranitz, J. M., Petanidou, T., Thorp, R. W., Wenner, A. M., Griswold, T. L. and Wells, H. (2009). Nectar secretion and bee guild characteristics of yellow star-thistle on Santa Cruz Island and Lesvos: where have the honey bees gone? *Uludag Bee Journal* 9, 109–121.

Barthell, J. F., Hranitz, J. M., Redd, J. R., Clement, M. L., Crocker, K. C., Becker, E. C., Leavitt, K. D., McCall, B., Mills-Novoa, M., Walker, C. M. et al. (2012). Observations on nectar availability and bee visitation at patches of yellow star-thistle and chasteberry on the northeast Aegean island of Lesbos (Greece). *Uludag Bee Journal* 12(2), 55–61.

Barthell, J. (2019). Foraging patterns of three carpenter bee species at chasteberry bushes in an Aegean Island ecosystem.

Baselga, A., Lobo, J. M., Svenning, J.-C., Aragón, P. and Araújo, M. B. (2012). Dispersal ability modulates the strength of the latitudinal richness gradient in European beetles. *Glob. Ecol. Biogeogr.* doi:10.1111/j.1466-8238.2011.00753.x.

Bartholomew, G. A. (1964). The roles of physiology and behaviour in the maintenance of homeostasis in the desert environment. *Symp. Soc. Exp. Biol.* 18, 7–29.

Bates, D., Mächler, M., Bolker, B. M. and Walker, S. C. (2015). Fitting linear mixed-effects models using *lme4*. *J. Stat. Softw.* 67, 1–48. doi:10.18637/jss.v067.i01.

Batra, S. W. T. (1966). Nesting behavior of *Halictus scabiosae* in Switzerland (Hymenoptera, Halictidae). *Insectes Sociaux* 13, 87–92.

Bikos, T. and Rammou, E. (2002). Beehives of the Aegean islands. *Bee World* 83(1), 5–13.

Bogert, C. M. (1949). Thermoregulation in reptiles, a factor in evolution. *Evolution* 3, 195–211.

Bogusch, P., Bláhová, E. and Horák, J. (2020). Pollen specialists are more endangered than non-specialised bees even though they collect pollen on flowers of non-endangered plants. *Arthropod-Plant Interactions* 14, 759–769.

Borrell, B. J. and Medeiros, M. J. (2004). Thermal stability and muscle efficiency in hovering orchid bees (Apidae: Euglossini). *J. Exp. Biol.* 207, 2925–2933.

Brand, N. and Chapuisat, M. (2012). Born to be bee, fed to be worker? The caste system of a primitively eusocial insect. *Frontiers in Zoology* 9(1), 1–9.

Buckner, M. A. and Danforth, B. N. (2022). Climate-driven range shifts of a rare specialist bee, *Macropis nuda* (Melittidae), and its host plant, *Lysimachia ciliata* (Primulaceae). *Global Ecology and Conservation* 37, e02180.

Burton, T., Ratikainen, I. I. and Einum, S. (2022). Environmental change and the rate of phenotypic plasticity. *Glob. Change Biol.* 28, 5337–5345.

Cameron, S. A., Whitfield, J. B., Hulslander, C. L., Cresko, W. A., Isenberg, S. B. and King, R. W. (1996). Nesting biology and foraging patterns of the solitary bee *Melissodes rustica* (Hymenoptera: Apidae) in northwest Arkansas. *J. Kans. Entomol. Soc.* 69, 260–273.

Cane, J. H. (1987). Estimation of bee size using intertegular span (Apoidea). *J. Kans. Entomol. Soc.* 60, 145–147.

Cane, J. H. and Neff, J. L. (2011). Predicted fates of ground-nesting bees in soil heated by wildfire: thermal tolerances of life stages and a survey of nesting depths. *Biol. Conserv.* 144, 2631–2636.

Cane, J. H. (2015). Landscaping pebbles attract nesting by the native ground-nesting bee *Halictus rubicundus* (Hymenoptera: Halictidae). *Apidologie* 46, 728–734.

Cartagena, S. M. (2021). Inter-individual variation in circadian rhythm across species of halictid bees: characterizing varying clocks and motif classification. Master's thesis, University of Puerto Rico, Rio Piedras, Puerto Rico.

- Casey, L. M., Rebelo, H., Rotheray, E. and Goulson, D. (2015). Evidence for habitat and climatic specializations driving the long-term distribution trends of UK and Irish bumblebees. *Diversity and Distributions* 21(8), 864–875.
- Corbet, S. A., Fussell, M., Ake, R., et al. (1993). Temperature and the pollinating activity of social bees. *Ecol. Entomol.* 18(1), 17–30.
- Chen, I. C., Hill, J. K., Ohlemüller, R., Roy, D. B. and Thomas, C. D. (2011). Rapid range shifts of species associated with high levels of climate warming. *Science* 333(6045), 1024–1026.
- Clusella-Trullas, S., Blackburn, T. M. and Chown, S. L. (2011). Climatic predictors of temperature performance curve parameters in ectotherms imply complex responses to climate change. *Am. Nat.* 177, 738–751.
- Crist, T. O. and Williams, J. A. (1999). Simulation of topographic and daily variation in colony activity of *Pogonomyrmex occidentalis* (Hymenoptera: Formicidae) using a soil temperature model. *Environ. Entomol.* 28, 659–668.
- Cossins, A. (2012). *Temperature Biology of Animals*. Springer, Dordrecht.
- Danforth, B. N., Conway, L. and Ji, S. (2003). Phylogeny of eusocial *Lasioglossum* reveals multiple losses of eusociality within a primitively eusocial clade of bees (Hymenoptera: Halictidae). *Syst. Biol.* 52, 23–36.
- Davis, M. B. and Shaw, R. G. (2001). Range shifts and adaptive responses to Quaternary climate change. *Science* 292(5517), 673–679.
- Davis, F. W., Synes, N. W., Fricker, G. A., McCullough, I. M., Serra-Diaz, J. M. and Franklin, J. et al. (2019). LiDAR-derived topography and forest structure predict fine-scale variation in daily surface temperatures in oak savanna and conifer forest landscapes. *Agric. For. Meteorol.* 269–270, 192–202.
- Deutsch, C. A., Tewksbury, J. J., Huey, R. B., Sheldon, K. S., Ghalambor, C. K., Haak, D. C. et al. (2008). Impacts of climate warming on terrestrial ectotherms across latitude. *Proc. Natl Acad. Sci. USA* 105, 6668–6672.
- Dimopoulos, T., Helfenstein, J., Kreuzer, A., Mohr, F., Sentas, S., Gianellis, R. and Kizos, T. (2023). Different responses to mega-trends in less favorable farming systems. Continuation and abandonment of farming land on the islands of Lesbos and Lemnos, Greece. *Land Use Policy* 124, 106435.
- Dobzhansky, T. (1950). Evolution in the tropics. *Am. Sci.* 38, 209–221.
- Dunham, A. E. (1993). Biotic interactions and global change. In: Kareiva, P. M., Kingsolver, J. G. and Huey, R. B. (eds), *Biotic Interactions and Global Change*. Sinauer Associates, Sunderland, MA, pp. 95–119.
- Dunn, T. and Richards, M. H. (2003). When to be social: interactions among environmental constraints, incentives, guarding, and relatedness in a facultatively social carpenter bee. *Behav. Ecol.* 14, 417–424.
- Eickwort, G. C., Eickwort, J. M., Gordon, J., Eickwort, M. A. and Weislo, W. T. (1996). Solitary behavior in a high-altitude population of the social sweat bee *Halictus rubicundus* (Hymenoptera: Halictidae). *Behav. Ecol. Sociobiol.* 38, 227–233.
- Eliopoulos, P. A. and Stathas, G. J. (2003). Temperature-dependent development of the koinobiont endoparasitoid *Venturia canescens* (Gravenhorst) (Hymenoptera: Ichneumonidae): effect of host instar. *Environ. Entomol.* 32, 1049–1055.
- Fabre, J. H. (1915). *Bramble-bees and others*. Dodd, Mead & Co., New York, 450 pp. (Translated by de Mattos).
- Fawcett, F. V. E., Minckley, R. L. and Neff, J. L. (2019). *The Solitary Bees: Biology, Evolution, Conservation*. Princeton University Press.
- Forrest, J. and Miller-Rushing, A. J. (2010). Toward a synthetic understanding of the role of phenology in ecology and evolution. *Philos. Trans. R. Soc. B* 365, 3101–3112.
- Forrest, J. R. K. and Chisholm, S. P. M. (2017). Direct benefits and indirect costs of warm temperatures for high-elevation populations of a solitary bee. *Ecology* 98, 359–369.
- Forrest, J. R. K., Cross, R. and CaraDonna, P. J. (2019). Two-year bee, or not two-year bee? How voltinism is affected by temperature and season length in a high-elevation solitary bee. *Am. Nat.* 193, 560–574.
- Frazier, M. R., Huey, R. B. and Berrigan, D. (2006). Thermodynamics constrains the evolution of insect population growth rates: “Warmer is better.” *Am. Nat.* 168, 512–520.
- Fritts, T. H. and Rodda, G. H. (1998). The role of introduced species in the degradation of island ecosystems: a case history of Guam. *Annu. Rev. Ecol. Syst.* 29, 113–140.
- Gathmann, A., Greiler, H.-J. and Tscharnke, T. (1994). Trap-nesting bees and wasps colonizing set-aside fields: succession and body size, management by cutting and sowing. *Oecologia* 98, 8–14.
- Gilbert, N. and Raworth, D. A. (1996). Insects and temperature—a general theory. *Can. Entomol.* 128, 1–13.
- Gil-Tapetado, D., Ferrari, A., Ronchetti, F., Polidori, C. and others (2024). Distribution widening of a ground-nesting social bee across Europe favored by climate change and urban setting. *Apidologie* 55, 1–20.
- Gogala, A. (1991). Initiation of new nests in a social bee, *Halictus scabiosae* Rossi (Hymenoptera: Halictidae). *Opusc. Zool. Flumin.* 67, 1–7.
- González, V. H., Patton, R., Plascencia, M., Girşgin, A. O., Çakmak, I. et al. (2018). High levels of tolerance between nestmates and non-nestmates in the primitively eusocial sweat bee *Halictus scabiosae* (Rossi) in Turkey (Hymenoptera: Halictidae). *Insectes Sociaux* 65, 339–343.
- Gonzalez, V. H., Osborn, A. L., Brown, E. R., Pavlick, C. R., Enríquez, E., Tscheulin, T., Petanidou, T., Hranitz, J. M. and Barthell, J. F. (2020). Effect of pan trap size on the diversity of sampled bees and abundance of bycatch. *J. Insect Conserv.* 24, 409–420.
- Gonzalez, V. H., Hranitz, J. M., Percival, C. R., Pulley, K. L., Tapsak, S. T., Tscheulin, T., Petanidou, T. and Barthell, J. F. (2020). Thermal tolerance varies with dim-light foraging and elevation in large carpenter bees (Hymenoptera: Apidae: Xylocopini). *Ecol. Entomol.* 45, 688–696.
- Gonzalez Betancourt, V. H. (2023). From Colombia to Lesbos: Experiences with Bee Research and Island Life. *Am. Entomol.* 69, 38–39.
- Gonzalez, V. H., Herbison, N., Robles-Pérez, G., Panganiban, T., Haefner, L., Tscheulin, T., Petanidou, T. and Hranitz, J. (2024). Bees display limited acclimation capacity for heat tolerance. *Biol. Open* 13, bio060179.
- Gonzalez, V. H., Rancher, W., Vigil, R., Garino-Heisey, I., Oyen, K., Tscheulin, T., Petanidou, T., Hranitz, J. M. and Barthell, J. F. (2024). Bees remain heat tolerant after acute exposure to desiccation and starvation. *J. Exp. Biol.* 227, jeb249216.
- Gould, S. J. (1978). *Ever Since Darwin: Reflections in Natural History*. Burnett, London.
- Grafton-Cardwell, E. E., Gu, P. and Montez, G. H. (2005). Effects of temperature on development of vedalia beetle, *Rodolia cardinalis* (Mulsant). *Biol. Control* 32, 473–478.
- Greenleaf, S. S., Williams, N. M., Winfree, R. and Kremen, C. (2007). Bee foraging ranges and their relationship to body size. *Oecologia* 153, 589–596.
- Gregory, S. and Wright, I. (2005). Creation of patches of bare ground to enhance the habitat of ground-nesting bees and wasps at Shotover Hill, Oxfordshire, England. *Conserv. Evid.* 2, 139–141.
- Harmon-Threatt, A. (2020). Influence of nesting characteristics on health of wild bee communities. *Annu. Rev. Entomol.* 65, 39–56.
- Hartig, F. (2024). DHARMA: residual diagnostics for hierarchical (multi-level / mixed) regression models. R package version 0.4.7. Available at: <https://CRAN.R-project.org/package=DHARMA>
- Helmuth, B., Kingsolver, J. G. and Carrington, E. (2005). Biophysics, physiological ecology, and climate change: Does mechanism matter? *Annu. Rev. Physiol.* 67, 177–201.
- Hirata, M., Cronin, A. L., Kidokoro, M. and Azuma, N. (2005). Spatio-temporal variation of colony structure and eusociality level of the Japanese sweat bee *Lasioglossum (Evylaeus) duplex* (Hymenoptera: Halictidae). *Ecol. Res.* 20, 75–79.
- Hoffmann, W. A., Schroeder, W. and Jackson, R. B. (2002). Positive feedbacks of fire, climate and vegetation and the conversion of tropical savanna. *Geophys. Res. Lett.* 29, 2052.
- Hoffmann, A. A., Chown, S. L. and Clusella-Trullas, S. (2013). Upper thermal limits in terrestrial ectotherms: how constrained are they? *Funct. Ecol.* doi:10.1111/j.1365-2435.2012.02036.x.
- Hogendoorn, K. and Zammit, J. (2001). Benefits of cooperative breeding through increased colony survival in an allodapine bee. *Insectes Soc.* 48, 392–397.
- Houston, A. I., McNamara, J. M. and Hutchinson, J. M. C. (1993). General results concerning the trade-off between gaining energy and avoiding predation. *Philos. Trans. R. Soc. Lond. B* 341, 375–397.
- Howe, R. W. (1967). Temperature effects on embryonic development in insects. *Annu. Rev. Entomol.* 12, 15–42.

- Huang, M. J., Hughes, A. C., Xu, C. Y., Miao, B. G., Gao, J. et al. (2022). Mapping the changing distribution of two important pollinating giant honeybees across 21,000 years. *Global Ecology and Conservation* 39, e02282.
- Huey, R. B., Hertz, P. E. and Sinervo, B. (2003). Behavioral drive versus behavioral inertia in evolution: a null model approach. *Am. Nat.* 161, 357–366.
- Huey, R. B., Deutsch, C. A., Tewksbury, J. J., Vitt, L. J., Hertz, P. E., Alvarez Perez, H. J. et al. (2009). Why tropical forest lizards are vulnerable to climate warming. *Proc. Biol. Sci.* 276, 1939–1948.
- Huey, R. B. and Stevenson, R. D. (1979). Integrating thermal physiology and ecology of ectotherms: A discussion of approaches. *Am. Zool.* 19, 357–366.
- Jackson, S. T. and Overpeck, J. T. (2000). Responses of plant populations and communities to environmental changes of the late Quaternary. *Paleobiology* 26, 194–220.
- Jeanne, R. L. and Morgan, R. C. (1992). The influence of temperature on nest site choice and reproductive strategy in a temperate zone *Polistes* wasp. *Ecol. Entomol.* 17, 135–141.
- Johansson, M. P., Quintela, M. and Laurila, A. (2016). Genetic divergence and isolation by thermal environment in geothermal populations of an aquatic invertebrate. *J. Evol. Biol.* 29, 1701–1712.
- Jones, J. C. and Oldroyd, B. P. (2006). Nest thermoregulation in social insects. *Adv. Insect Physiol.* 33, 153–191.
- Kamm, D. R. (1974). Effects of temperature, day length, and number of adults on the sizes of cells and offspring in a primitively social bee (Hymenoptera: Halictidae). *J. Kans. Entomol. Soc.* 47, 8–18.
- Kapheim, K. M., Bernal, S. P., Smith, A. R., Nonacs, P. and Weislo, W. T. (2011). Support for maternal manipulation of developmental nutrition in a facultatively eusocial bee, *Megalopta genalis* (Halictidae). *Behav. Ecol. Sociobiol.* 65, 1179–1190.
- Kearney, M., Shine, R. and Porter, W. P. (2009). The potential for behavioral thermoregulation to buffer “cold-blooded” animals against climate warming. *Proc. Natl Acad. Sci. USA* 106, 3835–3840.
- Kearney, M. R., Simpson, S. J., Raubenheimer, D. and Kooijman, S. A. L. M. (2013). Balancing heat, water and nutrients under environmental change: a thermodynamic framework. *Funct. Ecol.* doi:10.1111/1365-2435.12020.
- Kerr, J. T., Pindar, A., Galpern, P., Packer, L., Potts, S. G. et al. (2015). Climate change impacts on bumblebees converge across continents. *Science* 349(6244), 177–180.
- Kier, G., Kreft, H., Lee, T. M., Jetz, W., Ibisch, P. L., Nowicki, C., Mutke, J. and Barthlott, W. (2009). A global assessment of endemism and species richness across island and mainland regions. *Proc. Natl. Acad. Sci. USA* 106, 9322–9327.
- Kingsolver, J. G. and Huey, R. B. (1998). Evolutionary analyses of morphological and physiological plasticity in thermally variable environments. *Am. Zool.* 38, 545–560.
- Kingsolver, J. G. and Huey, R. B. (2008). Size, temperature, and fitness: three rules. *Evol. Ecol. Res.* 10, 251–268.
- Kougiouloutzis, K., Kaloveloni, A. and Petanidou, T. (2022). Assessing climate change impacts on Island bees: The Aegean Archipelago. *Biology* 11(4), 552.
- Knerer, G. (1967). Zur Bienenfauna Niederösterreichs: die Unterfamilie Halictinae. *Zool. Anz.* 181, 82–117.
- Knerer, G. (1969). Synergistic evolution of halictine nest architecture and social behavior. *Can. J. Zool.* 47, 925–930.
- Knerer, G. (1980). Biology and social behaviour of bees of the genus *Halictus* Latreille (Hymenoptera: Halictidae). *Zool. Jahrb. Syst.* 107, 511–536.
- Knerer, G. (1992). The biology and social behaviour of *Evyllaes malachurus* (K.) (Hymenoptera: Halictidae) in different climatic regions of Europe.
- Knerer, G. and Plateaux-Quénu, C. (1967). Usurpation de nids étrangers et parasitisme facultatif chez *Halictus scabiosae* (Rossi) (Insecte Hyménoptère). *Insectes Soc.* 14(1), 47–50.
- Kukuk, P. F. (2002). Nest reuse in a communal halictine bee, *Lasioglossum (Chilalictus) hemichalceum* (Hymenoptera: Halictidae). *J. Kans. Entomol. Soc.* 75, 3–7.
- Lanner, J., Dubos, N., Geslin, B., Leroy, B., Hernández-Castellano, C. et al. (2022). On the road: anthropogenic factors drive the invasion risk of a wild solitary bee species. *Science of the Total Environment* 827, 154246.
- Larsson, F. K. (1991). Some take it cool, some like it hot: a comparative study of male mate searching tactics in two species of Hymenoptera (Colletidae and Sphecidae). *J. Therm. Biol.* 16, 45–51.
- Lienhard, A., Mirwald, L., Hötzel, T., Kranner, I. and Kastberger, G. (2010). Trade-off between foraging activity and infestation by nest parasites in the primitively eusocial bee *Halictus scabiosae*. *Psyche* 2010, 707501.
- Lin, N. (1964). Increased parasitic pressure as a major factor in the evolution of social behavior in halictine bees. *Insectes Soc.* 11, 187–192.
- Legewie, H. (1925). Zur Theorie der Staatenbildung. I. Teil: Die Biologie der Furchenbiene *Halictus malachurus* K. *Z. Morphol. Okol. Tiere* 5, 619–684.
- Lomolino, M. V., Riddle, B. R. and Whittaker, R. J. (2017). *Biogeography*. Oxford University Press, Oxford.
- Lyman, C. P. (1948). The oxygen consumption and temperature regulation of hibernating hamsters. *J. Exp. Zool.* 109, 55–78.
- Marshall, T. J., Holmes, J. W. and Rose, C. W. (1996). *Soil Physics*, 3rd edn. Cambridge University Press, Cambridge, UK.
- Marshall, L., Ascher, J. S., Whittaker, R. J., Orr, M. C., Hughes, A. C., Schrader, J., Weigelt, P., Kreft, H. and Vereecken, N. J. (2026). Global Patterns and Drivers of Bee Diversity and Endemism on Islands. *bioRxiv* 2026–02.
- Martínez-López, O., Koch, J. B., Martínez-Morales, M. A., Navarrete-Gutiérrez, D., Enríquez, E. and Vandame, R. (2021). Reduction in the potential distribution of bumble bees (Apidae: *Bombus*) in Mesoamerica under different climate change scenarios: conservation implications. *Global Change Biology* 27(9), 1772–1787. <https://doi.org/10.1111/gcb.15559>
- MacArthur, R. H. and Wilson, E. O. (1967). The theory of island biogeography. Princeton University Press.
- Melville, J. and Schulte II, J. A. (2001). Correlates of active body temperatures and microhabitat occupation in nine species of central Australian agamid lizards. *Austral Ecol.* 26, 660–669.
- Michener, C. D. (1979). Biogeography of the bees. *Ann. Missouri Bot. Gard.* 66, 277–347.
- Michener, C. D. (2007). *The Bees of the World*. Johns Hopkins University Press.
- Mikola, J., Virtanen, T., Linkosalmi, M., Vähä, E., Nyman, J., Postanogova, O. et al. (2018). Spatial variation and linkages of soil and vegetation in the Siberian Arctic tundra—coupling field observations with remote sensing data. *Biogeosciences* 15, 2781–2801.
- Mitchella, D. J. (1985). Host life-history variation in response to parasitism. *Parasitology* 90(1), 205–216.
- Miyayama, R., Maeta, Y. and Sakagami, S. F. (1999). Geographical variation of sociality and size-linked color patterns in *Lasioglossum (Evyllaes) apristum* (Vachal) in Japan (Hymenoptera: Halictidae). *Insectes Soc.* 46, 224–232.
- Nemésio, A., Silva, D. P., Nabout, J. C. and Varela, S. (2016). Effects of climate change and habitat loss on a forest-dependent bee species in a tropical fragmented landscape. *Insect Conservation and Diversity* 9(2), 149–160.
- Neokosmidis, L., Tschulin, T., Devalez, J. and Petanidou, T. (2018). Landscape spatial configuration is a key driver of wild bee demographics. *Insect Science* 25(1), 172–182.
- Nielsen, A., Steffan-Dewenter, I., Westphal, C., Messinger, O., Potts, S. G., Roberts, S. P. M., Settele, J., Szentgyörgyi, H., Vaissière, B. E., Vaitis, M. et al. (2011). Assessing bee species richness in two Mediterranean communities: importance of habitat type and sampling techniques. *Ecol. Res.* 26(5), 969–983.
- Noll, J. (1931). Untersuchungen über die Zeugung und Staatenbildung des *Halictus malachurus* Kirby. *Z. Morphol. Okol. Tiere* 23, 285–368.
- Norden, B. B. (1984). Nesting biology of *Anthophora abrupta* (Hymenoptera: Anthophoridae). *J. Kans. Entomol. Soc.* 57, 243–262.
- Nuttman, C. and Willmer, P. (2003). How does insect visitation trigger floral colour change? *Ecol. Entomol.* 28(4), 467–474.
- Ollerton, J., Winfree, R. and Tarrant, S. (2011). How many flowering plants are pollinated by animals? *Oikos* 120, 321–326.

- Packer, L.** (1983). The nesting biology and social organisation of *Lasioglossum (Evylaeus) laticeps* (Hymenoptera, Halictidae) in England. *Insectes Soc.* 30, 367–375.
- Packer, L. and Knerer, G.** (1985). Social evolution and its correlates in bees of the subgenus *Evylaeus* (Hymenoptera; Halictidae). *Behav. Ecol. Sociobiol.* 17, 143–149.
- Packer, L.** (1986). Multiple-foundress associations in a temperate population of *Halictus ligatus* (Hymenoptera; Halictidae). *Can. J. Zool.* 64, 2325–2332.
- Packer, L. and Knerer, G.** (1986b). The biology of a subtropical population of *Halictus ligatus* Say (Hymenoptera: Halictidae). *Behav. Ecol. Sociobiol.* 18, 363–375.
- Packer, L., Sampson, B., Lockerbie, C. and Jessome, V.** (1989). Nest architecture and brood mortality in four species of sweat bee (Hymenoptera: Halictidae) from Cape Breton Island, Nova Scotia, Canada. *Can. J. Zool.* 67, 2864–2870.
- Packer, L.** (1998). A phylogenetic analysis of western European species of the *Lasioglossum leucozonium* species-group (Hymenoptera: Halictidae): sociobiological and taxonomic implications. *Can. J. Zool.* 76, 1611–1621.
- Pauly, A., Pesenko, Y. and Radchenko, V.** (2016). Les *Halictus* Latreille, 1804 d'Europe et du Bassin Méditerranéen. Atlas Hymenoptera. Available at: <http://www.atlashymenoptera.net/page.aspx?id=70>
- Paxton, R. J., Kukuk, P. F. and Tengö, J.** (1999). Effects of familiarity and nestmate number on social interactions in two communal bees, *Andrena scotica* and *Panurgus calcaratus* (Hymenoptera: Andrenidae). *Insectes Soc.* 46, 109–118.
- Petanidou, T. and Lamborn, E.** (2005). A land for flowers and bees: studying pollination ecology in Mediterranean communities. *Plant Biosyst.* 139, 279–294.
- Plateaux-Quénu, C.** (1993). Modalités de la socialisation chez les Halictinae (Hymenoptera, Halictidae). I: Biologie des Halictinae. *L'Année Biologique* 32, 183–204.
- Plateaux-Quénu, C., Plateaux, L. and Packer, L.** (2000). Population-typical behaviours are retained when eusocial and non-eusocial forms of *Evylaeus albipes* (F.) (Hymenoptera: Halictidae) are reared simultaneously in the laboratory. *Insectes Soc.* 47, 263–270.
- Plateaux-Quénu, C.** (2008). Subsociality in halictine bees. *Insectes Soc.* 55, 335–346.
- Polidori, C., Rubichi, A., Barbieri, V., Trombino, L. and Donegana, M.** (2010). Floral resources and nesting requirements of the ground-nesting social bee, *Lasioglossum malachurum* (Hymenoptera: Halictidae), in a Mediterranean semiagricultural landscape. *Psyche* 2010, 851947.
- Polidori, C., García-Gila, J., Blasco-Aróstegui, J. and Gil-Tapetado, D.** (2021). Urban areas are favouring the spread of an alien mud-dauber wasp into climatically non-optimal latitudes. *Acta Oecologica* 110, 103678.
- Porter, S. D.** (1988). Impact of temperature on colony growth and developmental rates of the ant *Solenopsis invicta*. *J. Insect Physiol.* 34, 1127–1133.
- Potts, S. and Willmer, P. A. T.** (1997). Abiotic and biotic factors influencing nest-site selection by *Halictus rubicundus*, a ground-nesting halictine bee. *Ecol. Entomol.* 22, 319–328.
- Potts, S. G., Petanidou, T., Roberts, S., O'Toole, C., Hulbert, A. and Willmer, P.** (2006). Plant-pollinator biodiversity and pollination services in a complex Mediterranean landscape. *Biol. Conserv.* 129, 519–529.
- Potts, S. G., Imperatriz-Fonseca, V., Ngo, H. T., Aizen, M. A., Biesmeijer, J. C., Breeze, T. D., Dicks, L. V., Garibaldi, L. A., Hill, R. and Settele, J.** (2016). Safeguarding pollinators and their values to human well-being. *Nature* 540, 220–229.
- Quénu, C.** (1957). Sur les femelles d'été de *Halictus scabiosae* (Rossi). *C. R. Acad. Sci. Paris* 244, 1073–1076.
- Rajver, D., Šafanda, J. and Dedeček, P.** (2006). Monitoring of air-ground temperature coupling and examples of shallow subsurface warming in Slovenia. *Geologija* 49, 279–293.
- Redding, T. E., Hope, G. D., Fortin, M. J., Schmidt, M. G. and Bailey, W. G.** (2003). Spatial patterns of soil temperature and moisture across subalpine forest-clearcut edges in the southern interior of British Columbia. *Can. J. Soil Sci.* 83, 121–130.
- Rehan, S. M., Schwarz, M. P. and Richards, M. H.** (2011). Fitness consequences of ecological constraints and implications for the evolution of sociality in an incipiently social bee. *Biol. J. Linn. Soc.* 103, 57–67.
- Richards, O. W.** (1953). *The Social Insects*. MacDonald, London.
- Richards, M. H. and Packer, L.** (1998). Demography and relatedness in multiple-foundress nests of the social sweat bee, *Halictus ligatus*. *Insectes Soc.* 45, 97–109.
- Richards, M. H.** (2000). Evidence for geographic variation in colony social organization in an obligately social sweat bee, *Lasioglossum malachurum* Kirby (Hymenoptera; Halictidae). *Can. J. Zool.* 78, 1259–1266.
- Richards, M. H., von Wettberg, E. J. and Rutgers, A. C.** (2003). A novel social polymorphism in a primitively eusocial bee. *Proc. Natl. Acad. Sci. USA* 100, 7175–7180.
- Ricklefs, R. E.** (2006). Evolutionary diversification and the origin of the diversity–environment relationship. *Ecology* 87, S3–S13.
- Roe, A.** (2024). Cross-pollinating ideas: what bee research in Greece taught me about global science collaboration. *Am. Entomol.* 70(4), 32–33.
- Roubik, D. W. and Michener, C. D.** (1980). The seasonal cycle and nests of *Epicharis zonata*, a bee whose cells are below the wet-season water table (Hymenoptera: Anthophoridae). *Biotropica* 12, 56–60.
- Sakagami, S. F. and Hayashida, K.** (1961). Biology of the primitive social bee, *Halictus duplex* Dalla Torre, III: activities in spring solitary phase. *J. Fac. Sci. Hokkaido Univ. Ser. VI Zool.* 14, 639–682.
- Schneider, C. A., Rasband, W. S. and Eliceiri, K. W.** (2012). NIH Image to ImageJ: 25 years of image analysis. *Nat. Methods* 9, 671–675.
- Schrader, J. et al.** (2024). Islands are key for protecting the world's plant endemism. *Nature* 634, 868–874.
- Schwarz, M. P., Richards, M. H. and Danforth, B. N.** (2007). Changing paradigms in insect social evolution: insights from halictine and allodapine bees. *Annu. Rev. Entomol.* 52, 127–150.
- Shebl, M.** (2012). The floral hosts of Japanese polylectic sweat bees of subgenus *Lasioglossum* (*Lasioglossum*) (Hymenoptera: Halictidae). *J. Apic. Res.* 51, 232–239.
- Shebl, M. A., Al Aser, R. M. and Ibrahim, A.** (2016). Nesting biology and seasonality of long-horned bee *Eucera nigrilabris* Lepeletier (Hymenoptera: Apidae). *Sociobiology* 63, 1031–1037.
- Sick, M., Ayasse, M., Tengö, J., Engels, W., Lübke, G. and Francke, W.** (1994). Host-parasite relationships in six species of *Sphecodes* bees and their halictid hosts: nest intrusion, intranidal behavior, and Dufour's gland volatiles (Hymenoptera: Halictidae). *J. Insect Behav.* 7, 101–117.
- Sih, A. and McCarthy, T. M.** (2002). Prey responses to pulses of risk and safety: testing the risk allocation hypothesis. *Anim. Behav.* 63, 437–444.
- Severns, P. M.** (2004). Creating bare ground increases presence of native pollinators in Kincaid's lupine seeding plots. *Ecol. Restor.* 22, 234–235.
- Smith, A. R., Weislo, W. T. and O'Donnell, S.** (2007). Survival and productivity benefits to social nesting in the sweat bee *Megalopta genalis* (Hymenoptera: Halictidae). *Behav. Ecol. Sociobiol.* 61, 1111–1120.
- Soucy, S. L.** (2002). Nesting biology and socially polymorphic behavior of the sweat bee *Halictus rubicundus* (Hymenoptera: Halictidae). *Ann. Entomol. Soc. Am.* 95, 57–65.
- Stone, G. N.** (1994). Activity patterns of females of the solitary bee *Anthophora plumipes* in relation to temperature, nectar supplies and body size. *Ecol. Entomol.* 19, 177–189.
- Stearns, S. C.** (1976). Life-history tactics: a review of the ideas. *Q. Rev. Biol.* 51, 3–47.
- Strohm, E. and Linsenmair, K.** (1998). Temperature dependence of provisioning behaviour and investment allocation in the European beewolf *Philanthus triangulum* F. *Ecol. Entomol.* 23, 330–339.
- Strumwasser, F.** (1960). Some physiological principles governing hibernation in *Citellus beecheyi*. *Bull. Mus. Comp. Zool.* 124, 285–320.
- Takkis, K., Tscheulin, T., Tsalkatis, P. and Petanidou, T.** (2015). Climate change reduces nectar secretion in two common Mediterranean plants. *AoB Plants* 7, plv111.
- Tattersall, G. J., Sinclair, B. J., Withers, P. C., Fields, P. A., Seebacher, F., Cooper, C. E. and Maloney, S. K.** (2012). Coping with thermal challenges: physiological adaptations to environmental temperatures. *Compr. Physiol.* 2, 2151–2202.

1681	Tipton, J. (2006). Aristotle's fish: The case of the kobios and phucis. <i>Perspect. Biol. Med.</i> 49, 369–383.	1737
1682	Truebano, M., Fenner, P., Tills, O., Rundle, S. D. and Rezende, E. L. (2018). Thermal strategies vary with life history stage. <i>J. Exp. Biol.</i> 221, jeb171629.	1738
1683		1739
1684	Tscheulin, T., Petanidou, T., Zografou, C. and Pantelis, K. (2008). Invading the Mediterranean: study of <i>Solanum elaeagnifolium</i> , a successful invasive plant from Central America. In: <i>Earth Conference</i> . University of the Aegean, Mytilini.	1740
1685		1741
1686		1742
1687	Tscheulin, T., Neokosmidis, L., Petanidou, T. and Settele, J. (2011). Influence of landscape context on the abundance and diversity of bees in Mediterranean olive groves. <i>Bull. Entomol. Res.</i> 101, 557–564.	1743
1688		1744
1689	Ulrich, Y., Perrin, N. and Chapuisat, M. (2009). Flexible social organization and high incidence of drifting in the sweat bee, <i>Halictus scabiosae</i> . <i>Mol. Ecol.</i> 18, 1791–1800.	1745
1690		1746
1691		1747
1692	Vandame, R., Morand, S., Colin, M. and Belzunces, L. (2002). Parasitism in the social bee <i>Apis mellifera</i> : quantifying costs and benefits of behavioral resistance to <i>Varroa destructor</i> mites. <i>Apidologie</i> 33(5), 433–445.	1748
1693		1749
1694		1750
1695	Vogt, J. T., Smith, W. A., Grantham, R. A. and Wright, R. E. (2003). Effects of temperature and season on foraging activity of red imported fire ants (Hymenoptera: Formicidae) in Oklahoma. <i>Environ. Entomol.</i> 32, 447–451.	1751
1696		1752
1697	Wcislo, W. T. and Cane, J. H. (1996). Floral resource utilization by solitary bees (Hymenoptera: Apoidea) and exploitation of their stored foods by natural enemies. <i>Annu. Rev. Entomol.</i> 41, 257–286.	1753
1698		1754
1699		1755
1700	Weissel, N., Mitesser, O., Liebig, J., Poethke, H.-J. and Strohm, E. (2006). The influence of soil temperature on the nesting cycle of the halictid bee <i>Lasioglossum malachurum</i> . <i>Insectes Soc.</i> 53, 390–398.	1756
1701		1757
1702	Weserling, J. and Tschardtke, T. (1995). Habitat selection of bees and digger wasps: experimental management of plots. <i>Commun. Ger. Soc. Gen. Appl. Entomol.</i> 9, 697–701.	1758
1703		1759
1704		1760
1705	West, E. S. (1952). A study of the annual soil temperature wave. <i>Aust. J. Sci. Res.</i> 5, 303–314.	1761
1706		1762
1707	Westrich, P. (1990). <i>Die Wildbienen Baden-Württembergs</i> , vols. 1–2, 2nd ed. Ulmer, Stuttgart, Germany.	1763
1708	Wickham, H. (2016). <i>ggplot2: Elegant Graphics for Data Analysis</i> . Springer-Verlag, New York.	1764
1709		1765
1710	Willmer, P. G. and Stone, G. N. (2004). Behavioural, ecological, and physiological determinants of the activity patterns of bees. <i>Adv. Study Behav.</i> 34, 347–466.	1766
1711		1767
1712	Whitfield, G. H. and Richards, K. W. (1992). Temperature-dependent development and survival of immature stages of the alfalfa leafcutter bee <i>Megachile rotundata</i> (Hymenoptera: Megachilidae). <i>Apidologie</i> 23, 11–23.	1768
1713		1769
1714	Woods, W. A., Heinrich, B. and Stevenson, R. D. (2005). Honeybee flight metabolic rate: does it depend upon air temperature? <i>J. Exp. Biol.</i> 208, 1161–1173.	1770
1715		1771
1716		1772
1717	Wuellner, C. T. and Jang, Y. (1996). Natural history of a ground-nesting solitary bee, <i>Crawfordapis luctuosa</i> (Hymenoptera: Colletidae). <i>J. Kans. Entomol. Soc.</i> 69, 211–221.	1773
1718		1774
1719		1775
1720	Wuellner, C. T. (1999). Nest site preference and success in a gregarious, ground-nesting bee <i>Dieunomia triangulifera</i> . <i>Ecol. Entomol.</i> 24, 471–479.	1776
1721		1777
1722	Xu, M., Chen, J. and Brookshire, B. L. (1997). Temperature and its variability in oak forests in the southeastern Missouri Ozarks. <i>Clim. Res.</i> 8, 209–223.	1778
1723		1779
1724	Yagi, N. and Hasegawa, E. (2011). Both cooperation and nest position improve larval survival in the sweat bee, <i>Lasioglossum (Evylaeus) baleicum</i> . <i>J. Ethol.</i> 29, 63–67.	1780
1725		1781
1726		1782
1727	Yanega, D. (1988). Social plasticity and early-diapausing females in a primitively social bee. <i>Proc. Natl. Acad. Sci. USA</i> 85, 4374–4377.	1783
1728		1784
1729	Yanega, D. (1997). Demography and sociality in halictine bees (Hymenoptera: Halictidae). In: <i>The Evolution of Social Behavior in Insects and Arachnids</i> , p. 293. Cambridge University Press.	1785
1730		1786
1731	Zevgolis, Y. G., Kamatsos, E., Akriotis, T., Dimitrakopoulos, P. G. and Troumbis, A. Y. (2022). Estimating productivity, detecting biotic disturbances and assessing the health state of traditional olive groves, using nondestructive phenotypic techniques. <i>Sustainability</i> 14(1), 391.	1787
1732		1788
1733		1789
1734		1790
1735		1791
1736		1792

Table S1. Species-level comparison of chill-coma recovery and take-off performance traits in HSC (*H. scabiosae*) and LMA (*L. malachurum*). Abbreviations: T_R = righting temperature, D_R = righting duration, T_T = take-off temperature, and D_T = take-off duration.

Species	ITD (mm)	T_R (°C)	D_R (s)	T_T (°C)	D_T (s)
HSC	0.340	29.8	185	32.8	505
LMA	0.206	26.4	131	31.2	468

Table S2. Daily variation in mean bee activity (mean number of bees observed per time interval) across sampling dates (15/4–24/4). Values represent mean activity per observation interval with associated standard deviation (SD) and sample size (n). Hourly analysis indicated peak activity between 11:00–12:00 h, with consistently elevated activity observed between 10:00–14:00 h across the study period. These temporal patterns are broadly consistent with previously reported foraging activity windows in *H. scabiosae* (Lienhard et al., 2010).

Date	Mean activity	SD	n
15/4	1.36	0.91	16
16/4	0.78	0.82	16
17/4	0.84	0.67	16
18/4	1.49	1.16	16
19/4	1.40	1.06	16
20/4	1.55	1.41	15
21/4	1.10	1.12	18
22/4	0.47	0.51	17
23/4	0.17	0.26	18
24/4	0.99	1.07	18

Table S3. Summary of focal behavioural metrics for HSC (*H. scabiosae*) and LMA (*L. malachurum*). Activity is expressed as the mean number of bees per observation interval. Guarding time is presented as a mean duration (seconds). Nest-entry and nest-exit frequencies are calculated as the mean number of events across five standardised time points (2, 4, 6, 8, and 10 minutes) within each observation interval; these values therefore represent standardised activity rates rather than instantaneous counts.

Species	Activity	Guarding time (s)	Nest-entry frequency	Nest-exit frequency
HSC	0.969	186.557	0.626	0.491
LMA	0.254	NA	0.307	0.245

Table S4. Linear mixed-effects models (LMMs) examining physiological and behavioural performance traits in HSC (*H. scabiosae*) and LMA (*L. malachurum*). Sampling date was included as a random intercept in all models. HSC was used as the reference level for species comparisons. Results are reported as regression coefficients (β), standard errors (SE), model degrees of freedom (df), t-statistics, and associated p -values.

Random intercept variance associated with sampling date was 1.148 for righting temperature, 0 for take-off temperature, 0.066 for righting duration, and 0 for take-off duration. The models for take-off temperature and take-off duration exhibited singular fits, indicating that sampling date contributed negligible variance in these cases.

Response variable	Predictor	β	SE	Test statistic	p
Righting temperature	LMA (vs HSC)	-2.488	0.876	$t = -2.84$	0.0078
	Ambient temperature	0.977	0.149	$t = 6.55$	< 0.001
Take-off temperature	LMA (vs HSC)	-2.100	1.115	$t = -1.88$	0.069
	Ambient temperature	1.413	0.234	$t = 6.03$	< 0.001
Righting duration (log-transformed)	LMA (vs HSC)	-0.422	0.214	$t = -1.98$	0.057
	Ambient temperature	0.033	0.036	$t = 0.90$	0.377
Take-off duration (log-transformed)	LMA (vs HSC)	0.003	0.251	$t = 0.01$	0.989
	Ambient temperature	0.058	0.053	$t = 1.09$	0.284

Table S5. Generalised additive models (GAMs) describing non-linear relationships between environmental predictors and behavioural activity (mean bee abundance). Smooth terms are reported using Effective Degrees of Freedom (EDF), approximate F -statistics, and associated p -values. EDF reflects the complexity of the fitted smooth term, with values close to 1 indicating approximately linear relationships and higher values indicating increasing non-linearity. Models were fitted using restricted maximum likelihood (REML; $n = 94$).

Predictor	EDF	F	p	Deviance explained (%)
Ambient air temperature	3.19	18.98	< 0.001	43.9
Shallow soil temperature	1.91	1.26	0.350	3.9
Middle soil temperature	2.27	4.22	0.007	13.5
Deep soil temperature	2.63	6.57	< 0.001	19.7

Table S6. Linear, linear mixed-effects, and generalised linear mixed-effects models linking environmental predictors to nest activity and abundance patterns in *H. scabiosae* (HSC) and *L. malachurum* (LMA). Sampling day and/or site were included as random effects in mixed-effects models. GLMM results are reported as log-odds with Wald z -test statistics; LM and LMM results are reported as regression coefficients with t -test statistics.

Response variable	Model	Predictor	β	SE	Test statistic	p
Focal occurrence models						
Guard presence (HSC)	GLMM	Shallow temperature (scaled)	-0.942	0.269	$z = -3.50$	< 0.001
Nest guarding duration (HSC, log)	LMM	Middle temperature (scaled)	0.506	0.166	$t = 3.04$	0.004
Abundance model (species-combined)						
Abundance (log)	LM	Species (LMA vs HSC)	-0.368	0.051	$t = -7.25$	< 0.001
		Ambient temperature	-0.028	0.056	$t = -0.51$	0.614
		Shallow temperature	0.318	0.083	$t = 3.85$	< 0.001
		Middle temperature	-0.289	0.103	$t = -2.81$	0.006
		Deep temperature	-0.045	0.077	$t = -0.59$	0.558
		Light intensity	0.242	0.051	$t = 4.74$	< 0.001

Table S7. Generalised additive models (GAMs) describing non-linear relationships between environmental variables and nest activity in *H. scabiosae* (HSC) and *L. malachurum* (LMA). Nest-entry and nest-exit frequencies were calculated as mean values across five observation time points per video (2, 4, 6, 8, and 10 minutes), and should therefore be interpreted as standardised activity rates rather than instantaneous counts. Smooth terms are reported with Effective Degrees of Freedom (EDF), approximate F -statistics, and associated p -values.

Response variable	Predictor	EDF	F	p
HSC nest-entry frequency	Surface temperature	2.14	3.54	0.024
HSC nest-exit frequency	Surface temperature	1.00	7.43	0.0078
HSC nest-exit frequency	Shallow temperature	1.00	4.46	0.038
LMA nest-entry frequency	Surface temperature	2.36	6.32	< 0.001
LMA nest-exit frequency	Ambient temperature	1.00	4.61	0.035

Table S8. Daily climatological summary for 15–24 April 2026 (Molyvos, Lesbos; 290 m a.s.l.). Mean air temperature (T_{mean}) over the period was 14.53 °C, and mean relative humidity (RH; calculated as the mean of daily maximum and minimum values) was 72.6%. Total rainfall was 17.4 mm, recorded on three days of the sampling period (15, 22, and 23 April). Data source: Meteo Aegean meteorological database (<http://meteo.aegean.gr>).

Day	T_{mean} (°C)	T_{max}	Tmax time	T_{min}	Tmin time	RH _{max}	RH _{min}	Rain (mm)
15	16.3	19.8	15:00	13.8	02:20	79	60	0.4
16	17.8	23.7	15:20	13.5	23:50	75	43	0.0
17	13.1	17.2	14:30	10.3	07:20	85	63	0.0
18	14.1	19.9	14:50	10.3	06:10	79	36	0.0
19	13.7	19.4	13:00	8.6	06:20	80	47	0.0
20	15.9	21.1	12:00	9.1	05:10	75	45	0.0
21	17.3	21.1	15:50	13.3	05:50	84	53	0.0
22	14.3	18.1	15:50	10.2	23:40	93	71	10.6
23	11.2	16.0	17:30	8.1	00:00	93	60	6.4
24	12.6	17.2	12:20	6.9	01:20	82	55	0.0

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Table S9. Multi-year April climatological summary (2003–2019) from the Agia Paraskevi meteorological station, the closest available long-term weather station to Skala Kallonis on Lesbos. Values are presented as monthly means (for April) for each year. Mean air temperature (T_{air}) across years was 14.6 °C, with mean maximum ($T_{\text{air,max}}$) and minimum ($T_{\text{air,min}}$) values of 15.3 °C and 13.9 °C, respectively. Mean relative humidity (RH) was 67.0%, with mean maximum (RH_{max}) and minimum (RH_{min}) values of 70.5% and 63.9%. Mean soil temperature (T_{soil}) was 14.9 °C, with mean maximum and minimum values of 15.8 °C and 14.3 °C, respectively. Data source: Meteo Aegean meteorological database (<http://meteo.aegean.gr>).

Year	T_{air}	$T_{\text{air,max}}$	$T_{\text{air,min}}$	RH	RH_{max}	RH_{min}	T_{soil}	$T_{\text{soil,max}}$	$T_{\text{soil,min}}$
2019	14.4	14.9	13.8	75.4	78.8	72.3	14.0	14.3	14.0
2018	18.3	19.0	17.6	62.8	66.3	59.4	14.0	14.3	14.0
2017	15.1	15.8	14.4	64.2	67.7	60.5	14.9	15.2	14.7
2016	17.6	18.3	16.9	64.4	68.0	60.8	16.2	16.4	16.0
2015	13.3	14.0	12.7	68.5	72.1	65.1	12.1	12.3	12.0
2014	15.7	16.3	15.1	73.2	76.8	69.5	15.0	15.2	14.8
2013	16.0	16.7	15.3	64.8	68.3	61.0	15.5	15.7	15.3
2012	14.5	15.1	13.9	73.0	76.0	69.7	15.1	15.3	14.9
2011	11.4	11.9	10.9	73.9	76.2	71.4	12.7	14.0	11.5
2010	14.3	15.0	13.8	61.8	64.9	58.6	14.4	14.5	14.2
2009	13.5	14.1	12.9	61.9	65.2	58.4	13.2	13.4	13.1
2008	14.2	14.8	13.7	69.8	72.6	66.7	14.2	14.4	14.1
2006	14.3	15.0	13.7	55.2	58.4	51.9	15.8	15.9	15.6
2005	13.0	13.6	12.3	53.4	56.5	49.9	14.5	14.7	14.3
2004	13.1	13.7	12.5	65.6	68.9	62.3	14.7	14.9	14.5

Table S10. Annual summer maximum air temperatures (T_{max}) at the Agia Paraskevi meteorological station, the closest long-term climate station to Skala Kallonis on Lesbos. Values represent the highest monthly maximum temperature recorded during summer months (June–August, depending on the year). Across the study period (2003–2019), mean summer T_{max} was 28.0 °C. Data source: Meteo Aegean meteorological database (<http://meteo.aegean.gr>).

Year	Month of T_{max}	T_{max} (°C)
2003	August	27.4
2004	July	26.0
2005	July	27.0
2006	July	25.7
2007	June	28.6
2008	August	27.4
2009	July	27.0
2010	August	29.0
2011	July	27.0
2012	July	32.1
2013	August	29.0
2014	August	29.0
2015	August	29.4
2016	July & August	29.0
2017	July	28.3
2018	August	29.0
2019	August	29.0

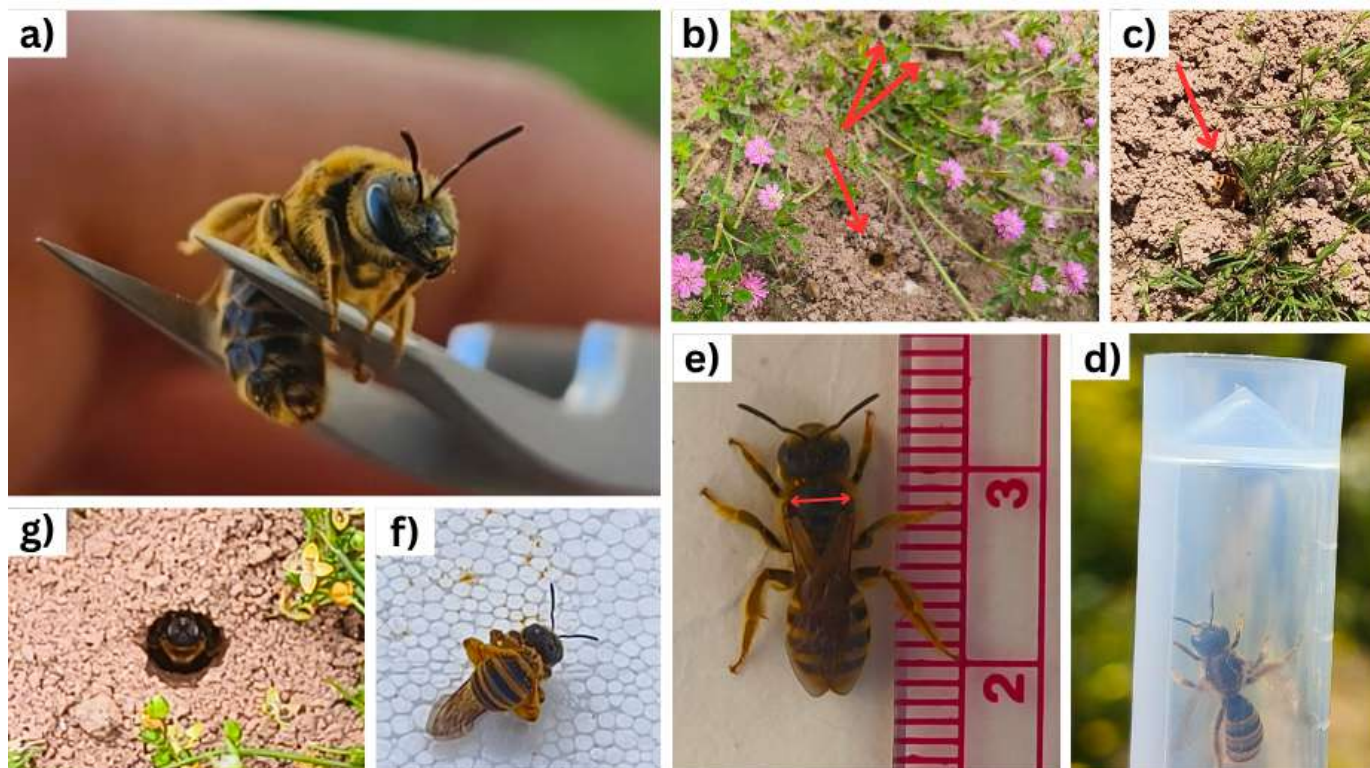


Fig. S1. Overview of field observations and chill-coma recovery assays in *H. scabiosae*. (a) Adult *H. scabiosae* profile. (b) Nesting aggregation of *H. scabiosae* within vegetation dominated by Persian clover (*Trifolium resupinatum*) at the field site; black lines indicate open nest entrances. (c) Black line indicating an individual *H. scabiosae* entering a nest within the aggregation. (d) Field capture of a *H. scabiosae* individual using an insect net, followed by transfer to a plastic vial for chill-coma recovery assays. (e) Chill-coma induced individual positioned dorsal side upward alongside a scale for ImageJ-based morphometric analyses; red double-headed line indicates ITD. (f) Chill-coma recovery posture of an individual placed ventral side upward on an insulated surface during thermal assays; the curled abdominal posture shown here is characteristic of recovery behaviour in *H. scabiosae*. (g) Nest guarding behaviour in *H. scabiosae*, with an individual positioned at the nest entrance.



Fig. S2. *L. malachurum* during chill coma recovery assays. (a) Individual positioned ventral side down, placed adjacent to a scale for ImageJ-based estimation of ITD. (b) Individual during the chill coma recovery assay, having righted itself but not yet initiating flight.

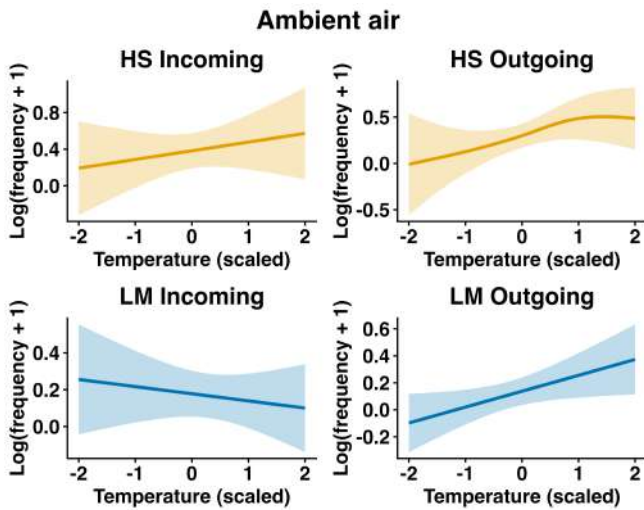


Fig. S3. Generalised additive model smooths showing the relationship between ambient air temperature and nest entry (incoming) and nest exit (outgoing) frequencies in *H. scabiosae* and *L. malachurum*. Shaded ribbons represent 95% confidence intervals.

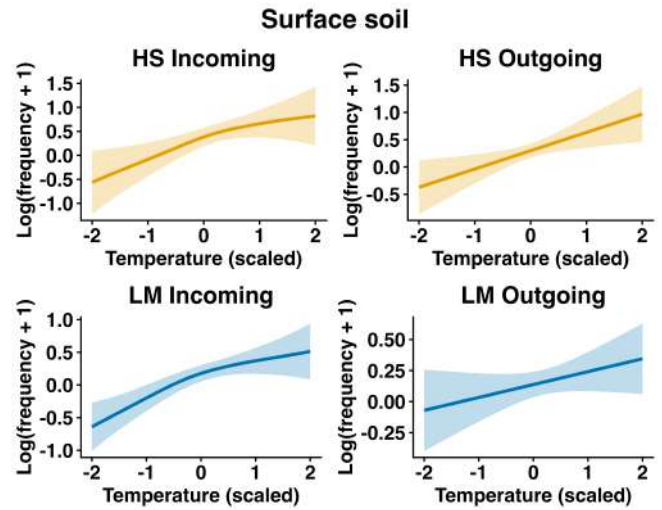


Fig. S4. Generalised additive model smooths showing the relationship between surface soil temperature and nest entry (incoming) and nest exit (outgoing) frequencies in *H. scabiosae* and *L. malachurum*. Shaded ribbons represent 95% confidence intervals.

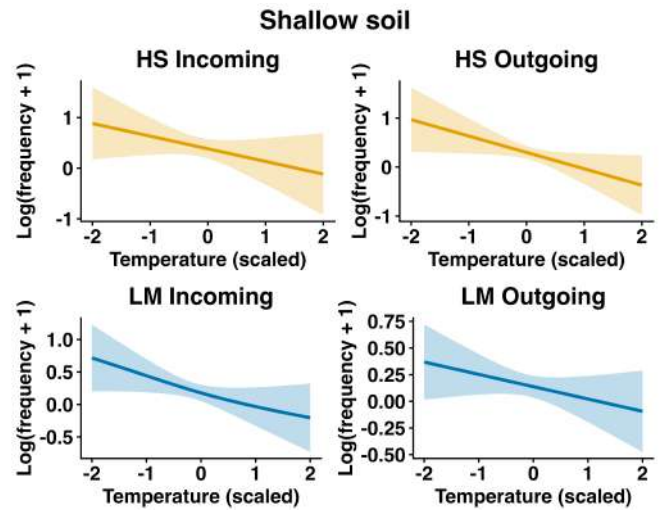


Fig. S5. Generalised additive model smooths showing the relationship between shallow soil temperature (2–3 cm depth) and nest entry (incoming) and nest exit (outgoing) frequencies in *H. scabiosae* and *L. malachurum*. Shaded ribbons represent 95% confidence intervals.

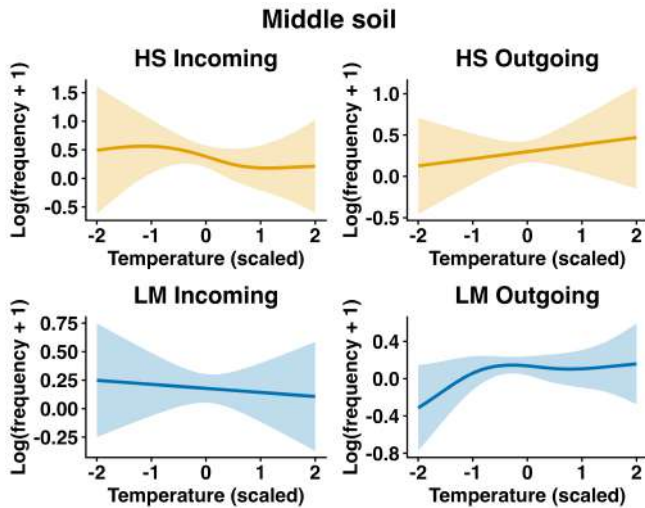


Fig. S6. Generalised additive model smooths showing the relationship between middle soil temperature (5 cm depth) and nest entry (incoming) and nest exit (outgoing) frequencies in *H. scabiosae* and *L. malachurum*. Shaded ribbons represent 95% confidence intervals.

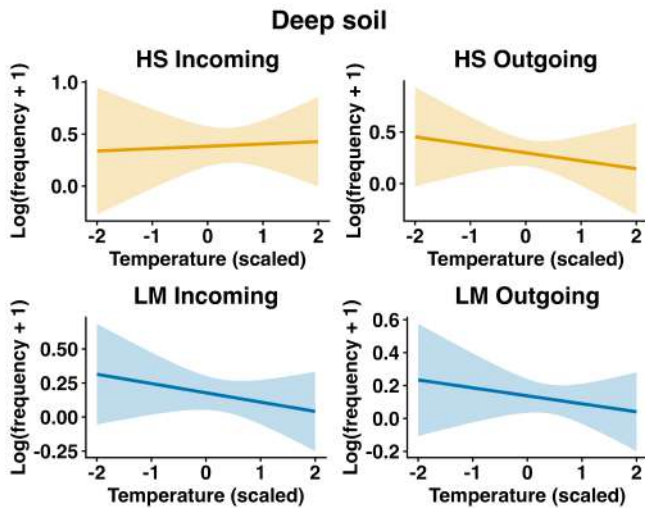


Fig. S7. Generalised additive model smooths showing the relationship between deep soil temperature (10 cm depth) and nest entry (incoming) and nest exit (outgoing) frequencies in *H. scabiosae* and *L. malachurum*. Shaded ribbons represent 95% confidence intervals.

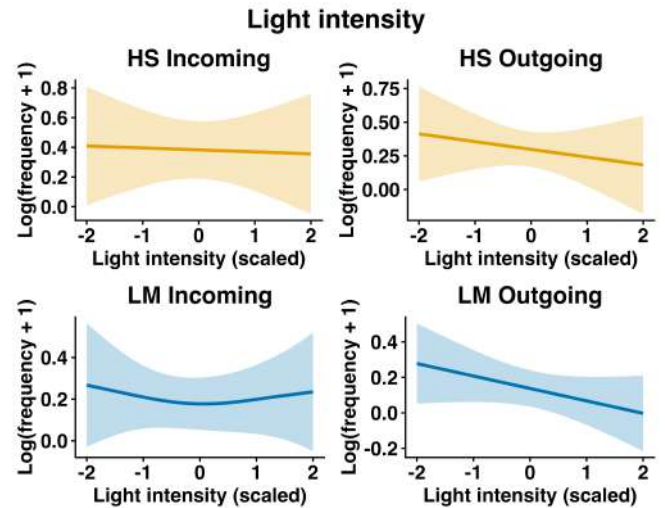


Fig. S8. Generalised additive model smooths showing the relationship between light intensity and nest entry (incoming) and nest exit (outgoing) frequencies in *H. scabiosae* and *L. malachurum*. Shaded ribbons represent 95% confidence intervals.

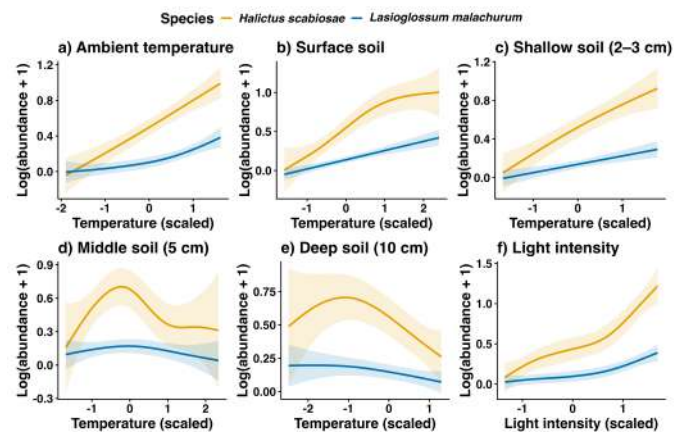


Fig. S9. Generalised additive model smooths showing abundance responses of *H. scabiosae* and *L. malachurum* across environmental gradients, including ambient air temperature, soil temperatures at multiple depths, and light intensity. Shaded ribbons represent 95% confidence intervals.

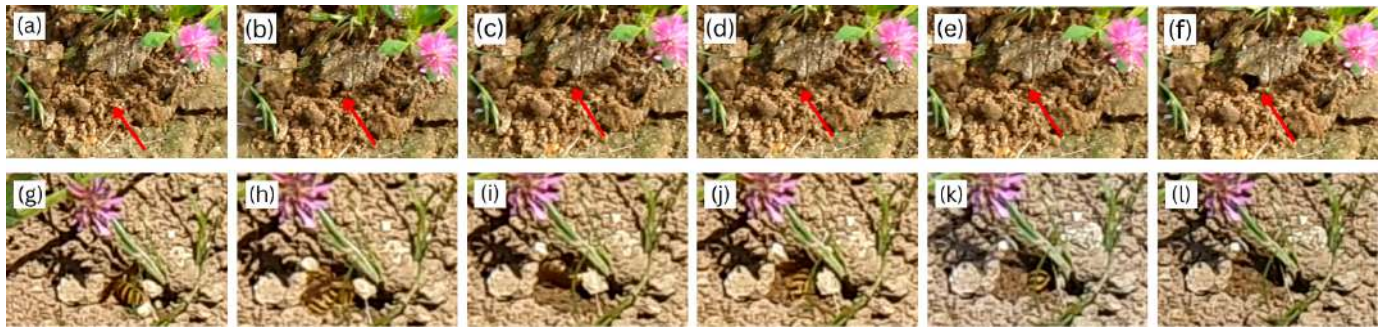


Fig. S10. Nest opening and new nest excavation behaviour in *H. scabiosae* at Site 2. (a–f) Nest opening behaviour recorded on 15 April during the 10:00–11:00 h observation period. Panels correspond to video timestamps measured from 10:00 h: (a) 08:03, (b) 09:46, (c) 09:52, (d) 11:56, and (e) 12:47, representing an approximately 4 min 44 s nest opening sequence. Red arrows indicate the location of the nest entrance. The individual occupying the nest entrance is visible in panels (c), (d), and (e). (g–l) Excavation of a new nest entrance in loose soil recorded on 19 April during the 13:00–14:00 h observation period. Panels correspond to video timestamps measured from 13:00 h: (g) 00:41, (h) 01:19, (i) 02:37, (j) 03:22, (k) 04:36, and (l) 04:47, representing an approximately 4 min 06 s excavation sequence. The excavating individual is visible in all panels except (l).

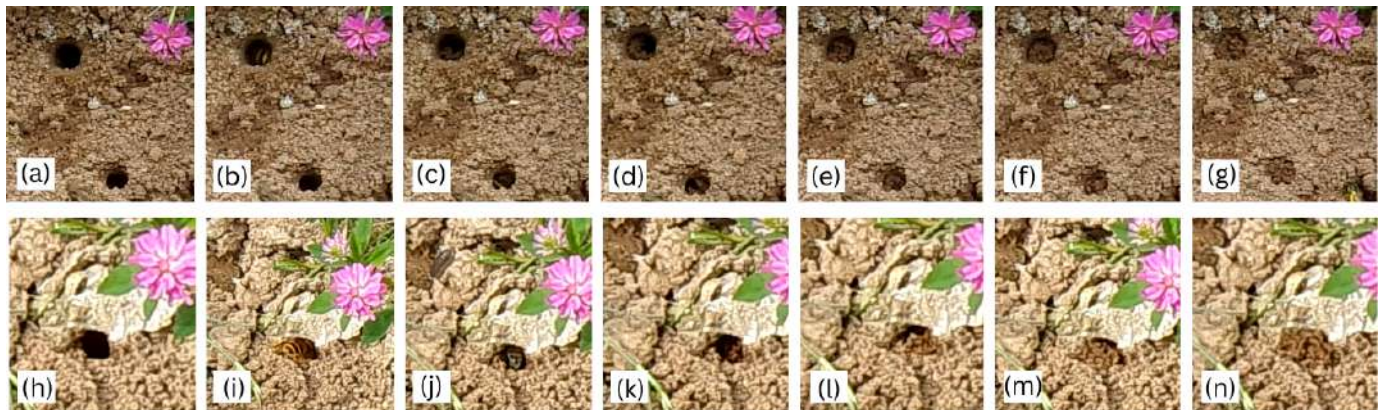


Fig. S11. Nest closing behaviour in *H. scabiosae* at Site 2. (a–g) Simultaneous closure of two nest entrances recorded on 17 April during the 13:00–14:00 h observation period. Panels correspond to video timestamps measured from 13:00 h: (a) 07:31, (b) 07:49, (c) 08:17, (d) 08:26, (e) 08:28, (f) 08:50, and (g) 09:23, representing an approximately 1 min 52 s nest closing sequence. (h–n) Nest closing behaviour recorded on 15 April during the 16:00–17:00 h observation period. Panels correspond to video timestamps measured from 16:00 h: (h) 00:36, (i) 01:21, (j) 02:20, (k) 04:35, (l) 05:18, (m) 06:33, and (n) 09:26, representing an approximately 8 min 50 s nest closing sequence.



Fig. S12. Changes in nest entrance status of *H. scabiosae* and *L. malachurum* nests within the common aggregation at Site 1 throughout a single day. Panels show the same field of view recorded at hourly intervals: (a) 10:00 h, (b) 11:00 h, (c) 12:00 h, (d) 13:00 h, and (e) 14:00 h. Red arrows indicate open or partially open nest entrances visible within the aggregation.

Reflections



Image 1: A photograph taken on the final day of fieldwork at my study site. Never much of a fan of selfies, but working alone in the field made occasional self-documentation feel necessary, both as proof of presence and as a memory of a place and time I know I will dearly miss in the years to come.

Coming to Lesbos was serendipitous, as I mentioned; it was never part of the grand plan, and yet here I was, doing terrestrial ecology on this most enchanting of islands. In many ways, it was different from Gran Canaria for me, principally because I was in the midst of company that felt much more wholesome and endearing than what I had experienced before, and that alone changes one's experiences and overall stay. This island hosts not only the usual Erasmus+ students, many of whom seem to arrive with the sole intention of amassing ever more blackout stories to later recount with gusto and pride, but also volunteers working at the refugee camp, people whose perspectives on life often felt profoundly different and whose experiences across the myriad organisations operating there made them an absolute pleasure to talk to and get to know. When I stepped into my fourth semester, I was already slightly late in starting my thesis, since I had decided to take yet another ecology course at RUG called *Island Biology*—and if one is familiar with my programme, this is not a course I could afford to miss. Thus, I was already under a mild time crunch when most of my peers had begun their thesis fieldwork. I felt as if a ticking time bomb was constantly hovering nearby, though in hindsight that pressure worked in my favour, forcing me to plan properly. The project, initially proposed as something grounded in chronobiology and sociality, gradually morphed

into something entirely different. However, the literature I had read before arriving on Lesbos was not wasted; instead, it became a springboard for thinking more broadly about activity rhythms in field conditions. Through discussions with chronobiologists at RUG, I came to realise that what I was proposing was certainly possible in principle, but within the constraints of a thesis timeframe it would have been infeasible and unnecessarily tedious. It would have required collecting bees across different levels of sociality for comparison, which is particularly tricky in a region where I had perhaps overestimated the existing natural history work. If I remember correctly, it was during the first online meeting with both Victor and Thomas that I realised the topic was already taking a different shape from what I had initially proposed. I was slightly taken aback at first, but given enough time to reflect before arriving on Lesbos, I realised that behavioural thermoregulation was not a bad direction either. I was keen to work in bee-behavioural ecology anyway, as it is the field I would like to pursue in my future doctoral studies. Gaining fieldwork experience in this area felt ideal, and it made more sense to install cameras in the field rather than work in a controlled lab environment, as I had originally proposed. My first in-person meeting with Thomas took place on 27 February, shortly after I arrived on the island and was still being drawn into its bee-guiling charm. I would admit that for the first two solid weeks, I had very little clear idea of what I would actually be doing here.

But gradually, through reading literature and digital ideation over emails with Thomas and Victor, I finally zoned in on two parts of my thesis: one behavioural ecology component, which would form the bulk of the work, and a second supplementary part involving physiological measurements of thermal limits to support trends observed in the first. It is funny to look back and see how much this kept changing and refining over time, because during my very first trip to the Malemi Hotel with Moritz, we were very happy to have found what I sincerely believed were the nests we were looking for, having never seen a ground-nesting bee nest before—though I am now fairly certain they were abandoned ant nests. I was at such a naive phase, almost an amateur with bees, that even honeybees looked bee-zarre to me, let alone other bee species that I could scarcely bee-gin to identify. I stared through my binoculars as they carried pollen in their little pollen baskets, and I was bee-wildered to realise that I had never observed a bee like this before, and that my binoculars were equally good at observing insects as birds; it was all far more overwhelming than I had expected, because insects are literally everywhere. When I was later told by Thomas that the actual study site was outside the hotel premises, I still stuck to my instincts and believed they might have missed these nests in previous years, especially since it seemed so convenient to me that they were within the hotel grounds. I thought it would make everything easier,



Image 2: (A) Overview of the multi-camera setup used during behavioural sampling across both field sites. Cameras included Thanasis' action camera for focal observations at Site 1 and my iPad for wide-angle coverage at the same site. At Site 2, Sveta's DSLR was used for focal observations and my GoPro for wide-angle coverage. Michael's handheld camcorder was used for opportunistic recordings of the field sites and surrounding behaviours. An additional camera angle was initially recorded using my phone but later replaced with Michael's GoPro due to the need to preserve the phone for other field purposes; this dataset was ultimately not used due to time constraints and redundancy with the primary GoPro recordings, particularly given the highly multi-angled distribution of nest entrances at Site 2. Michael's second GoPro was used to record the chill coma recovery assays. In total, 7–8 devices were in rotation, making battery management, charging logistics, and SD card limitations (often only 8 GB, requiring frequent swapping to prevent overwriting) a daily but demanding challenge that kept me constantly prepared for things going wrong. (B) Camera setup used at Field Site 1 during behavioural ecology sampling. Thanasis' action camera (visible in white) was positioned for focal recordings of selected nest entrances, while my iPad was used for wide-angle behavioural observations across the wider nesting aggregation. Note the discarded wiring and metal fencing strewn around the setup.



Image 3: (A) Chill coma recovery assay being conducted by me and Isabella on a honeybee individual. Isabella recorded body temperatures and recovery durations while I held the thermal camera. The icebox used for cold exposure is visible on the left, alongside the insect net and cotton balls used to seal plastic vials after bee collection. (B) Close-up of surface temperature measurements being taken from a bee positioned ventral side upwards during righting duration and body temperature recordings. (C) Experimental setup used for the chill coma recovery assay. All equipment was placed inside a thermally insulated chamber (a polystyrene box) used to minimise external temperature fluctuations during trials. The setup includes two stopwatches: one used to record the duration individuals spent in the icebox, with checks initially conducted minute-by-minute and later at 30-second intervals once movements subsided (icebox not shown). A thermocouple was used to measure ambient temperature, while a thermal camera (not switched on in this configuration) was intended for surface temperature measurements. A second stopwatch was used to record trial durations. Flight recovery was scored only when individuals were able to initiate true flight outside the chamber, excluding brief or incomplete “mock flights,” to maintain a consistent metric across both bee species.



Image 4: One of the first potential nests I initially sent to Victor, which I believed to be *H. scabiosae*. In retrospect, the nest entrance appears excessively surrounded by twigs and debris and was most likely an unused ant nest located within the hotel premises.

as I would not have to lug a 30 kg battery through tall vegetation every day, and since the cameras also needed to be transported daily because I could not leave them outside due to the risk of rain, it all felt more manageable. When I stayed at the hotel in March (25–30) in Skala Kallonis, eagerly waiting to begin preliminary data collection, Thomas and his PhD student Antigone helped me set up the cameras they had used in previous seasons. Although the setup seemed extremely cumbersome on the surface, requiring daily transport through dense vegetation without an assistant despite my attempts to recruit several juniors and Erasmus students, the real issue emerged when we inspected the site Thomas had in mind and found that the soil was waterlogged and there were no nests in sight. We were all disappointed, as it looked completely different compared to the dry summer conditions when the landscape was barren, and for me it was especially disheartening, since I had been hoping to begin sampling during this week of fieldwork. I decided to rig all six cameras at night while speaking to my sister to calm my nerves as panic built up, and I was genuinely proud when I managed to get everything working, although that moment quickly turned against me the next morning when I attempted to record what I think were “ant” nests, still believing they were bee nests, and discovered that the small monitor used to check camera angles and focus was not functioning. At that point, frustration, dis-bee-lief, worry, and panic all set in at once; I was anxious because not only did I not have any confirmed nests to record, as I had never observed bees entering the sites I had selected, but I also had no working camera system even if I

did. I video-called Thomas and Antigone, and Thomas decided to order a replacement system, similar to the existing rear-view car camera monitor, but it would only arrive on Monday, while I was originally due to leave on Tuesday. I therefore decided to use the remaining time for natural history scouting around the site, and that was when luck finally struck during a wandering survey of the unmanaged pasture behind the swimming pool fence of the Malemi Hotel, where I encountered the first large nesting aggregations. At the time, I did not even have an insect net, so using only a plastic vial, and hoping I would not get stung, as I initially believed these sweat bees stung as well, I carefully collected individuals for identification using AI before sending them to Victor and Thomas. This entry is the first in my field journal and is quite reflective of what my first proper field day looked like:

26th March 2026; 8:44 am Inside hotel premises: No visible activity Outside hotel premises: Wet soil, vegetation, promising

10:40 am Figuring out how this monitor works. I have set up the cameras, but haven't started recording yet.

7:05 pm

Horrible morning. I'm not the most technologically challenged person; at times I'm surprisingly rather adept with such things. But this morning I came close to admitting defeat, bee-reft of words to express my astonishment. I still do not have the faintest idea why some equipment that worked perfectly yesterday refused to work today. The only explanation I can fathom is that I used the same battery plugs as input for the "small screen," which is really an antique reverse camera monitor for a car from the 1980s, and that the battery could not handle the multiple connected systems running from it simultaneously.

Oh well. That at least forced me to explore the grounds a bit more, and lo and behold—I found *H. scabiosae* :)

Beautiful, shiny, and flying as though on acid—I initially thought it was a rabid wasp. ChatGPT told me it was a hoverfly. Gemini reacted with ardour, maintaining that it is *H. scabiosae*. ChatGPT failed. Having found enough nests for four cameras (originally there were six, but two had broken wires that could not be soldered back on), I was satisfied after all. My thesis did not sink; the ship is still afloat, though desperately in need of repair. The water is out for now.

After activity dwindled into oblivion around 3 pm, I ordered a heavy meal at Skala Kallonis and watched Balkan terrapins while my phone battery nearly died. I went back to check on the bees around 6 pm and was rudely driven out by pestilent hordes

of swarming mosquitoes. Now seated on this rather large bed with clean sheets, the air conditioner relegated to a heater, and a dinner buffet awaiting me, I feel as close as one can get to a privileged researcher from a first-world country conducting “extremely tedious and gruelling” fieldwork in whatever remains of the wilderness in a Mediterranean second-world country.

The only stark difference is that I still do not know whether I am paying for this out of my own pocket or not. I really hope the personal travel grant responds affirmatively tomorrow.

Worried and a bit lonesome,
Amey



Image 5: The unmanaged pasture meadow surrounding the nesting sites during the early phase of fieldwork. Initially, the nesting areas were dominated by Persian clovers, but over the following weeks the landscape changed rapidly, gradually becoming bee-sieged by yellow and white flowering patches due to blooming dandelions, chamomiles, and *Raphanus raphanistrum* (wild radish). Toward the later stages of sampling, scarlet pimpernels also began appearing near the nesting sites, making the meadow feel as though it transformed subtly every few days throughout the season.

And it did work out in the end—I managed to secure the personal travel grant after writing a proposal for this thesis:

Dear Members of the Review Committee,

I am writing to apply for a personal travel grant to support the expenses associated with the fieldwork for my master’s thesis. I am a second-year student in the Islands and Sustainability Erasmus Mundus master’s programme at the Faculty of Spatial Sciences, University of Groningen. I am currently based on the Greek island of Lesbos, where I am conducting my master’s thesis research on bee chronobiology and behavioural



Image 6: The very first nesting aggregation I encountered at the field site. After pushing through a dense thicket of grass and vegetation, this sudden barren patch of exposed soil appeared almost unexpectedly, and it was the moment I finally realised what the nesting aggregations actually looked like in the field. The soil seemed perfect: sandy, bare without vegetation, and dry, with countless *L. malachurum* nests scattered across the patch, which immediately gave me hope that the field season might actually work out after all.



Image 7: The first site where I observed multiple *H. scabiosae* nests clustered together, alongside numerous individuals actively foraging and collecting pollen and nectar from Persian clovers.



Image 8: Initial Site 2, located approximately five metres from Site 1, appeared to function as an extension of the main nesting aggregation and contained several additional isolated nests. At the time, I speculated that these may have represented separate nesting aggregations with different foundresses, and I sent these observations to Victor on 26 March. This site later became important during fieldwork, as I used Michael's camcorder here to record opportunistic behaviours and collect B-roll footage for a mini-documentary summarising my thesis fieldwork during the 10-minute observation periods conducted throughout the main sampling phase between 15–24 April.



Image 9: My final Field Site 2, which consisted of a mound-like “village” of nesting aggregations dominated by *H. scabiosae* and *L. malachurum*, alongside several other bees including species from the genera *Chelostoma* and *Andrena*. During the evenings, the site gradually became shaded by the adjacent wall, and I initially suspected that this evening shadow may have influenced activity patterns and possibly even the timing of nest-closing behaviour.



Image 10: This was one of the first moments when I realised that the bees had only recently begun excavating and establishing their nests, as indicated by freshly dug earth and the surrounding turret of loose sand (a tumulus). Activity peaked around midday, with the nesting aggregation surrounded by ants, bee flies, beetles, pill bugs, and numerous other arthropods that I could scarcely identify, possibly including brood parasites and kleptoparasites. I also observed, for the first time, female bees cautiously protruding either their heads or abdomens from nest entrances whenever another insect or nestmate approached too closely. By around 15:00, when I sent this image to Victor on 26 March, activity had already declined substantially.



Image 11: The red line indicates the route Victor used to access the back of the hotel and the approximate area where nesting aggregations had been located during previous summers when he conducted observations there. The purple markings represent the locations I initially believed to be the study sites I had found. In reality, these were entirely incorrect; the actual nesting aggregations described in the manuscript were situated much closer to the swimming pool and adjacent pathways. For a considerable time, however, this mistaken mental map persisted in my head, which in hindsight feels like a fitting reminder of just how disoriented and visually overwhelmed one can be at the very beginning of fieldwork. Sent to Victor on 27 March.

thermoregulation in affiliation with the University of the Aegean. At present, I am living in the city of Mytilene, which is approximately 43 km from my field site near Lake Kalloni, in the area adjacent to the Malemi Hotel. Previous studies in this location have been conducted by Dr. Victor Gonzalez (University of Kansas) and my supervisor, Dr. Thomas Tscheulin (University of the Aegean). Their work identified nesting aggregations of *Halictus scabiosae*, a species belonging to the Halictidae bee family, in the vicinity of the Malemi Hotel.

Halictus scabiosae is an obligately eusocial bee species that often reuses the same nests each year to form colonies typically consisting of 6–120 individuals. In some populations (for example in Switzerland), colonies show distinct castes and hierarchy, with one breeding female and several worker-forager females. As the season progresses, the filial generation (the queen's offspring) replaces the original workers after they are driven out of the nest. This process transforms the colony from an initially semi-social structure with multiple breeding females into a primitively eusocial colony characterised by overlapping generations, cooperative brood care, and reproductive division of labour. Displaced females may drift and establish nearby, resulting in nesting aggregations that make this species particularly suitable for behavioural monitoring.

For my thesis, I plan to deploy six cameras near these nests for two consecutive days each week. After each sampling period, I will return to Mytilene to review the footage and

ensure that the recordings were successful before the next week of data collection. I intend to begin fieldwork around 25th April, when female bees typically begin emerging and becoming active, and continue until the end of June. My thesis defence, which will take place in Gran Canaria (my mobility island), is scheduled for 30th June. Because of this schedule, I will need to travel to Kalloni each week for two days of fieldwork. The field site is relatively remote and the nearest available accommodation is the Malemi Hotel, which is located very close to the nesting area, effectively within the hotel's grounds. Staying there allows easy access to the site and the staff are also familiar with our research and help us store equipment safely. The cost of accommodation is €30 per night. Over a two-month period, this amounts to approximately €60 per week, €240 per month, and €480 in total. In addition, I estimate about €150 for food during the two months and approximately €80 for bus transportation between Mytilene and Kalloni. Altogether, a grant of around €650 would allow me to cover these essential fieldwork expenses.

I have planned to spread my sampling across multiple weeks rather than conducting it over a short continuous period. This approach helps reduce the influence of short-term weather conditions that might otherwise confound behavioural observations. Collecting data across several weeks will also allow me to capture subtle seasonal transitions from spring to early summer, which may influence bee behaviour. The camera recordings will later be analysed by scoring behavioural activity in 10–30 minute intervals (the exact methodology is still being refined). In addition to recording nest-entry and nest-exit frequencies, I will estimate the number of individuals active around each nest and approximate foraging bout durations where possible. I will also record environmental variables such as temperature, humidity, and light intensity (to determine whether nests are shaded or exposed). Another component of the analysis will involve documenting the presence of parasites or predators, including bee flies, cuckoo wasps, gold wasps, ichneumon wasps, *Sphecodes* bees, and velvet ants. A key behaviour of interest in my study is nest-closing behaviour, which often peaks after midday when parasite activity increases and bee foraging activity declines. We aim to investigate whether this behaviour functions primarily as an anti-parasite strategy or as a form of behavioural thermoregulation. Summer temperatures in Lesvos can reach around 40 °C, so temporarily sealing the nest may help regulate internal nest conditions. Observing this behaviour earlier in the season, when parasite pressure is still relatively low, may help clarify whether thermoregulation plays an important role, or whether the behaviour is primarily anticipatory and defensive.

Due to the limited timeframe before my thesis defence, my research focuses on a single

well-documented site. Collecting sufficient replicates from one location will allow me to obtain robust data within the available time. Ideally, I would also survey additional sites with similar environmental characteristics (sandy or loamy soils, thermophilic vegetation, and nearby floral resources), but this is not feasible within the current timeframe. Previous studies have also focused on single nesting aggregations, which provides a useful basis for comparison. Within the site itself, I will still be able to examine subtle environmental variation, for example by comparing nests in shaded versus exposed locations. In addition to nest-closing behaviour, I may also document other behaviours such as nest guarding, in which the queen positions herself at the nest entrance to prevent parasites or predators from entering. Dr. Victor Gonzalez has already collected data on activity patterns during the summer months, and my research will complement these observations by focusing on earlier seasonal activity and particularly on nest-closing behaviour before parasite activity peaks.

Although I receive an Erasmus Mundus stipend, a personal travel grant would greatly help me cover the additional costs required for this fieldwork. This support would allow me to carry out consistent weekly observations and collect the data necessary to complete my thesis successfully. If awarded the grant, I would be happy to share documentation of the fieldwork and acknowledge this funding support in my thesis and related outputs.

Thank you very much for considering my application. I really appreciate this opportunity and the time you have taken to review my application. I eagerly look forward to your response!

Sincerely,

Amey Danole

I think this was the first instance of me writing an actual proposal, not a mock one, for a real topic I would be working on. It is interesting to look back at the timelines mentioned here: quite ambitious, and the expenses I had estimated are still broadly close to the actual ones, although some, such as transportation, were way off, and the comparative analysis part of the story had not even been conceived yet. It is a shame to look back and realise that I could not afford to spend much more time at the site to obtain longitudinal data as I had originally proposed, since budget constraints, and later, during data analysis I realised quite harshly, time constraints, would not allow me to collect data over too many days. Additionally, the original plan to record parasites and predators near the site was not very successful because, although I had good camera quality and resolution in the end, there was simply so much happening that quantifying everything within the limited analysis timeframe seemed nearly impossible. But I am

rushing ahead; let us go back to my March stay, when the only humans I spoke to were at night, when a German tourist group hunting orchids around the island gathered for dinner. The rest of the time, I spoke with the countless African and South American parrots, rabbits, cockerels, squirrels, and South American coatis that the hotel owner kept on the premises. Through my days of observation, this week became instrumental to the questions I would later develop during the remainder of my thesis, and I am almost glad the cameras did not work and that I did not end up studying ants in despair and desperation instead of bees, which had been my plan B.

At 8:59 am on 27 March, I observed a bee-fly hovering over a *H. scabiosae* nest, witnessed through binoculars as it repeatedly attempted to flick its larvae while a single guarding individual tried to repel it, or at least that is how I interpreted it at the time, with some help from Gemini. It was also the first time I noticed *L. malachurum* lurking around the nesting site (confirmed by Victor on 27 March), and this was when I realised for the first time how different the nesting sites looked in the morning compared to the evening. My field journal entry at 9:23 am on 27 March reads: "The nests stand out like bullet holes. Each species marring the earth every which way, unique and architecturally incomprehensible." At 11:18 am, after the briefest of rain showers, I wrote: "But a lot of the 'bullet holes' are smoothed out now and it is really difficult to tell if there are nests here in the first place." I had a lot of free time between observation bouts, and when I needed a break from constantly looking through the binoculars and writing, especially poetically, it helped pass the time. I immediately noticed how important light intensity was, as bees would come out of their nests during brief bursts of sunlight when clouds moved aside, for example at 9:37 am: "whenever there is a moment of sunny respite, I see some activity in the bees." By 11:18 am, I was already considering a comparative idea after observing several small bees becoming active shortly after brief rainfall, which made me wonder whether these species might have different thermal tolerance limits, with some potentially emerging under lower light conditions, whereas *H. scabiosae* might require a threshold temperature before becoming active. I decided to record videos on my iPad and phone simply to practice analysis, which turned out to be a good decision because this was when I realised how good those recordings were. Being horrible at identifying plants (apologies, Theophrastus) and suffering from chronic plant blindness, I attempted a basic plant survey of the site and recorded insect taxa as well, which quickly became a herculean task due to the sheer diversity within such a small patch of land. I was also very naive at the time, under the belief that bees carrying pollen in the morning might somehow be different species, as their intensely yellow appearance seemed to bee-lie what they actually were, and it is almost laughable how overwhelmed I was at first, looking at this entirely new microcosmic

world of insects.

I had great trouble with the iButtons that were used because they were only compatible with Windows, and since I have a Mac, I struggled to find a Windows computer or laptop in order to start them recording. I had a similar issue with BORIS, as only a beta version of behavioural observation software was available for Mac, but I eventually decided against using it, since it added more time than it saved. The behaviours I was analysing were simple enough to go through directly from video and note down quickly, and in many recordings multiple nests appeared in the same frame, so I had to rely on the human eye rather than a software tool that would have required constant correction. The hotel staff were kind enough to let me use their computer, which was so old and worn out that I spent over two hours trying to get it to work, but to no avail, and they had to make other arrangements the next day by replacing the monitor and cleaning the CPU, which had accumulated eons of dust. Eventually, I was able to get the devices running and began collecting mock data as well. It is funny to realise that the thing that most defined my thesis (soil temperature) was the one I paid least attention to at the start; I initially thought the soil depth profile would simply add another interesting dimension rather than become the central theme of the first part of my thesis. But I am rushing ahead again.

In my early March days, I noticed many crickets sun-bathing on the dark turrets or tumuli excavated by the bees, and I thought that perhaps these darker structures allowed them to heat up faster than the surrounding soil. Whenever I saw bumblebees in the field, alongside the perpetual swooping swallows during their mating season, chirruping and moving in groups over the fields, I felt reassured, because it meant *H. scabiosae* would likely be active that day—since if bumblebees could heat up, so could *H. scabiosae*. I also noticed that many nests were embedded within vegetation, particularly Persian clover and buttercup patches, which seemed more protected from rain. I initially thought that later in the season they might shift toward more exposed nests in areas less influenced by vegetation in order to gain more heat, although I now realise that as summer progresses most vegetation disappears anyway and conditions become extremely hot and almost unbearable. Some iButton Hygrochron and Thermochron data loggers did fail, but this did not deter me, because I believed I had enough alternatives and initially assumed I would not need all of them, just one or two per site. Oh, how wrong I was. On 29 March (Sunday) at 15:39, I wrote: “My dad told me to get something precautionary for bee stings, which I agree with. I do seem to run the risk of clashing with a foraging bee.” This is important because it was then that I looked up what I should do in the unfortunate event of being stung. On the first day

of my fieldwork, I was indeed stung twice—once on my arm, which was quite painful, and another time when a bee entered my shoe while I was walking through vegetation, which sounds worse than it actually was. For the first sting, I panicked, dropped the camera I was holding, and rushed to my hotel room (since it happened right outside my room rather than at the field site). I then tried desperately to remove the stinger using a credit card-like object; having not found my debit card, I grabbed something random, most likely the curtain. Seeing no visible stinger, I simply washed the area with soap and water, and I was fine. I was very relieved not to be allergic, and after that day my fear decreased significantly. After returning to Mytilene, I decided to take matters into my own hands since it was my own thesis, and began trying to acquire action cameras by borrowing from friends, Erasmus students, or locals, or even renting them, because I believed they were far superior in quality and far less cumbersome than the CCTV-style cameras whose wires could easily come loose if not handled carefully and whose focus could be affected by the slightest movement. After asking around, I obtained two cameras, which was enough for me at the time since I also had my phone and iPad, and I later decided to invest in a GoPro to get a wide-angle view. However, I never anticipated that Michael would end up lending me three of his cameras during fieldwork, two of which I used to record my chill coma recovery assays and to film myself performing them, simply as memories and for the mini-documentary I had in mind. In the end, everything worked out on that front as well, despite how much I panicked at the beginning when no one was responding to my requests and messages.

When I used to sit in Thomas' lab, which was a shared space with another professor, I would share the space with Jelle, who gave me such valuable insight about *H. scabiosae* by telling me that it exhibited a form of competitive exclusion with *H. sexcinctus*, and that both species co-occur in the western Mediterranean. He also explained that *H. scabiosae* prefers plants from the family Asteraceae, and that one of the easiest ways to distinguish it from *H. sexcinctus* is the absence of mite pockets (often referred to as acarinarium), specialised morphological structures used to house and transport symbiotic mites. This made me worry for a while whether I was actually observing *H. sexcinctus* or *H. scabiosae*, and when he later confirmed after visiting my field site in April that it was indeed *H. scabiosae*, my joy knew no bounds, as it meant everything was on the right track. I tried recording environmental variables from the Agia Paraskevi weather station every day, but during fieldwork days it was sometimes down, and in the end there was so much of my own data that adding additional environmental datasets would have only made the thesis messier, so I decided against incorporating it. I also learned how to organise all of this data in the midst of fieldwork by writing explicit step-by-step instructions for myself to access websites, often in the simplest possible form,

such as “click here,” “scroll down here,” and similar notes, because I knew I would forget once actual fieldwork began, when there was simply too much on my mind. It was good that I anticipated this, because I did end up forgetting almost everything I had prepared in March during the transition into actual fieldwork.

“The mountains are clad in shadowy muslins, seductive as the clouds part, their bare interior reflects a million different rays.” — 28 March, 1:41 pm.

15 April: 9:46 am (day 1 of sampling): “Taking my first sample soon. A bit nervous”—after Johnny helped set up my cameras and tripod and we tried out different angles to record the previous day’s setup, which really boosted my confidence. Having someone to throw ideas around with made a big difference, and he did an excellent job guiding me through different configurations. I remember us even using our shadows to check whether the tips of the shadows appeared in the video frame as a rough way of estimating the field of view. I really owe him for helping me feel comfortable on the first day of fieldwork. I started recording from 10 am because the previous day we had noticed activity around 9:40 am. However, from the seventh day onwards, when Victor asked me how I knew for certain that nothing happened around 9 am on other days, I had no answer. So I started getting up earlier and eventually realised that activity actually began in the faintest form around 8:40 am. From then on, starting recording at 9 am made more sense, effectively adding an extra hour of field time from day seven onwards. By that point, I had grown into the routine and did not mind it at all. At the end of field days I was usually completely exhausted, and after a swim in the pool, some exercise, and playing with the two sweetest newly parent dogs—Hannah and Frodo—and their puppies, who were often alone in their large enclosures shared with African emus, exotic geese, and ducks, followed by some solo football, I would quickly eat dinner, go for a walk with others to Skala Kallonis bee-neath skies where clouds formed fantastically contoured shapes in the moonlight over the stolidly still nearby mountains, and then crash on my bed without thinking about anything or looking at my phone for long. In hindsight, these were easily the best two weeks I have had on Lesvos so far, hands down.

16 April; 10:03 am (day 2 of sampling): “No activity as yet. It is a bit more chilly today compared to yesterday. Hopefully it does not rain. Even the honeybees are not out yet, as they are usually active for most of the day, staying out quite late in the evening and emerging early before everyone else in the morning. Yesterday I saw an interesting behaviour: grasshoppers displaying atop nest mounds. Swallows and mosquitoes are out. Everything is very quiet on the soil. Not a single insect in sight. Mornings like these, beautiful as they may be, replete with bird chirrups and the fresh scent of mist-

laden mountains, scare me for the beings nestled snug in the soil, as it feels cold and treacherous to venture out.”

17 April (day 3 of sampling)—I wrote the reflections for field day 1 on field day 3. I had initially planned to write reflections for each day at the end of each field day, but I was too tired and preoccupied with other tasks, such as extracting environmental data from the previous day from the Meteo Aegean online meteorological database, and taking light intensity measurements after every 10-minute recording session at both sites using a lux meter app on my phone, both of which I ultimately did not use in my analysis. It is funny to think that in the moment these tasks feel crucial, yet in the bigger picture they often do not hold up, while the smallest and seemingly trivial details, like digging soil for the thermochrons using scissors and measuring their depths with a scale, end up mattering the most. Anyways, here are my reflections for field day 1:

“Scared and truly alone, starting out proved a lot gutsier than I thought. I started doubting myself and called Johnny to confirm the previous day’s onset activity. Reassured, I started the recordings with a ball in my stomach at 10 am. Looking back, 14 April was a marvellous day. The first camera angles we got were exceptionally good (for field site 2), and now, on my third day, time, overuse, and shifting conditions had already begun to strip every tripod setup of its original charm. I still think those first-day angles were somehow different: everything shimmered less, nest entrances were clear as daylight, and clarity reigned supreme. After two recordings, shocked and aghast, I saw Sveta’s camera die during the third session, and ever since (it has only been a day), I change it every second recording cycle because I assume it only lasts for two 10-minute sessions and I would not dare experiment with whether it survives a third, especially at this stage of fieldwork. I did test it earlier in the island’s lounge setting, where it lasted longer, but I had failed to factor in the intense heat of the sun beating down on the equipment. All in all, after two bee stings and two mistakes (one being forgetting to switch on, or rather check, Thanasis’ camera), it was disastrous, but still a learning curve.”

On my day 4 of sampling (18 April), Victor found *H. scabiosae* and *L. malachurum* on the hotel premises, right outside my room and the lab, which made it convenient to collect individuals and carry out trials. I was relieved to be able to work with these individuals there rather than disturb my behavioural sites for the chill coma recovery assay experiments, which Michael later helped me set up on the spot after I felt somewhat dejected when the critical thermal limits machine broke down. This idea came together quickly, and when he suggested it, I was glad it aligned so well with the behavioural component of my work, so that the two parts began to feel like

chapters of the same story told from different perspectives, ultimately integrating into a single framework, which felt quite exciting. And there were so many individuals of *L. malachurum* there that I did not need to worry about double-counting. I was more concerned about *H. scabiosae*, but then I remembered Victor's summer sites, and, as it turned out, new *H. scabiosae* nests had begun appearing there as well. This meant I could also procure additional individuals from those locations, as well as from the small site between my Field Sites 1 and 2 in times of desperate need. I began writing less and less in my field journal as the days went on, partly because I often left it on the ground near the sites, where it camouflaged so well with the soil that I sometimes forgot it was even there. My field sessions consisted of recording for the first 10 minutes of every hour from 9 to 6 pm, followed by 5-minute intervals in which I measured light intensity using a lux meter app on my phone, and then quickly running to collect bees for my chill coma recovery assays. Each trial took variable time, so achieving a sample size of 20 individuals per species required extreme speed and efficiency in my setup. For this reason I relied on voice recordings rather than written notes, narrating what was happening, how long each bee had been in chill coma, and whether chill coma had been induced, while simultaneously preparing equipment such as the thermal camera, thermocouples, GoPro, tweezers positioned near the measuring scale, and blank sheets for photographing and recording measurements. At the start, I recorded the bee weight (with vial), and I would then hold the thermal camera above the bee in chill coma and continue narrating what I observed, including duration, temperature, and recovery milestones such as righting and take-off, before finally recording the vial weight. It was extremely hectic, and I sometimes reduced lunch to 10–15 minutes because the activity window was short and I wanted to maximise sampling before 3 pm, after which chill coma recovery assays were no longer feasible due to the lack of active bees. In addition, the iButtons were not fully reliable due to ageing batteries, so I had to excavate them from the soil (5–10 cm deep depending on placement), extract the data, and reinstall them at their respective depths and positions, carefully recompact the soil to avoid disturbance. Initially, I was sceptical, as I thought all temperatures would be highly correlated across depths and that a single Thermochron per site would suffice. I am now glad I was proven wrong, as this allowed me to capture much more reliable temperature variation. On alternate days, this meant field sessions often lasted until 6:30 pm. Overall, this taught me how to commit fully to field time and accept delayed rewards, but I genuinely enjoyed it, especially being surrounded by equally passionate people like Michael, Victor, and Isabella, who helped me keep going. My final field day notes were quite sad; I was genuinely reluctant to leave, and I could not believe that this intense period would soon fade. I knew that returning to Mytilene would make

life feel dull again, and it did—but only for a while.

24 April; Friday (day 10 of sampling): “This has become my life now; alas, it all crescendos to an end. As all good things eventually do, it is coming to an end, and it will be really hard adjusting to life back in Mytilene. I have come to realise how much I like this phase of life: the rush that is persistently swelling, and how involved I am in things other than myself. I have come to see a family in Victor, Michael, and Isabella, and I will dearly miss them, most of all Michael. He made me feel heard after so long and tremendously helped me with equipment. I cannot believe today is the last day I spend here. Yesterday was especially miserable, full of rain and no activity, but I am still glad I could catch three individuals for my take-off study. I now have 10 good samples for each species, and I plan to do five more ideally for each species, though the best would be 20 individuals. My plan is to sample the other sites as best as I can today and to include individuals from my own site for the final day. Yesterday I had a funny moment when I was essentially waiting for a large cloud to pass, because bees come out of their nests on cold days only when there is a surge of sunlight breaking through, creating a brief glorious window that cannot be missed. It made me realise how much I take things for granted, and how in nature even the finest behaviours are governed by rhythms that stop for no one and proceed at their own pace, entirely bee-yond my control. I was at an *H. scabiosae* nest where Victor had sampled in previous years, and I saw an individual peep out of its nest and remain there.”

And thus, as unceremoniously as it had begun, my field journal ends here. Having learnt so much through this experience, the three pages I had initially dedicated to reflections have grown to twenty-two, and I still feel I have more to say. When I first began this master's, I came from a strongly quantitative background, and I still do, but I have learnt so much from my peers over the past two years in ways that have truly changed how I think about and practise science. Coming from a background where I had to secure scholarships to study and explore outside my country, I was deeply motivated to gain a place in a programme like this. I was initially rejected for the scholarship and placed on the waiting list, but eventually, after a withdrawal, I was offered the place and was able to join. The experience exceeded anything I had ever imagined. It helped me slow down just enough, look beyond my immediate field, and learn from other disciplines while still engaging deeply with my own. I have learnt to recognise my own prejudices and biases, and to accept that any study, however objective it may appear, carries a deeply subjective motivation that shapes it in subtle ways. There was also a considerable amount of subjectivity in this process, for example in estimating bee counts—whether there were 10–15 or 15–20 individuals—since I

simply did not have the time to count every single one. I had to rely on intuition in ways I never expected in a scientific project, but here I was, estimating abundances across time points, sometimes easily and sometimes with difficulty, yet consistently using the same margin of error; in this way, the broad behavioural component of the study gradually took shape. I panicked for the first time when I began data analysis, because after accumulating over 500 GB of data (and navigating several external hard drive failures), I finally had everything in place and analysis was all that remained. Four full days in the island's lounge, and working until 2 a.m. when I returned to my room each night, were required to push through this phase and move on to the results section. It was by far one of the hardest parts of the project, because although I could endure long hours in the field, staring at a screen for over 12 hours a day was something my eyes were neither prepared nor equipped for. Nevertheless, believing that one must sweat more in practice so that one bleeds less in battle, and with sticky notes lining my deadlines and people holding me accountable, I managed to complete it by a narrow margin and still find time to write this long reflection section, which I felt was necessary to do justice to what this programme has given me. I felt a sense of accomplishment unlike anything before, and the backdrop of bee-leaguered rejections from internships, research assistantships, and PhD applications did not deter me during this phase. This thesis taught me to think and act independently, with my supervisors serving as crucial fulcrums who guided my direction while leaving the details to me, for which I am deeply grateful, as I have come to understand the extent of my own independence. My limits were truly tested throughout this thesis, and I am glad they were, as I learnt a great deal from the experience. Although I had already encountered extensive ecology and evolution theory through my courses at RUG, each covering fascinating and diverse topics, it was only when I was in the field actively trying to understand why something was happening that I felt another part of my thinking truly activate for the first time, one that began identifying gaps in the project and considering how to improve or address them. This is why the future directions section of the thesis became so extensive, as many of these ideas emerged organically over time through doing the work itself, rather than from planning in advance.

Throughout these two years, I have been trained to think and act analytically, and I have learnt so much about the extraordinary world of ecology and evolutionary biology that I am certain a future in this field would make me happy. I have been shaped by so many people from different walks of life, and I carry a part of them wherever I go. The way I speak now, the way I choose my words more carefully, the way I listen, the way I carry myself, the way I get up after countless rejections, and the way things affect me have all changed. During my MITACS internship in Canada, Jacob, one of the master's students

there, told me to “pick the things worth getting upset about,” and I feel I have fully taken that on board through this master’s. I care more deeply about the things I truly care about, and I have stopped worrying about things I cannot control. I have learnt to navigate life—not just academically, but socially as well: adapting to places, to people, to the way they think, what I can learn and improve, and what I can safely ignore. I feel this programme has been a kind of training ground for life before stepping into a world of intense competition, where personality often seems to matter less than what is written on paper, whereas this programme has taught me to value the opposite. I am deeply grateful for the opportunity and the privilege of sharing this journey with an extraordinarily inspiring and warm cohort and endearing juniors, who made my time in Lesvos truly memorable, along with everything that Groningen, Gran Canaria, and Lesvos have taught me. That bee-ing said, I still have my last months in Lesvos ahead, and I will not let sentimentality take over and prevent me from enjoying them fully. I hereby write the final words I would write for my Islands and Sustainability master’s journey as a student. As an alumnus, I am sure I will keep coming back to this small island of ours, because that is what this programme and its experiences do: they brand you as an islander for life.